

Testing the Dual Representation Account of Memory for Trauma

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Abstract

The finding that a visuospatial intervention can decrease involuntary memories following a laboratory analogue of trauma, without altering voluntary memory, is a crucial source of support for the Dual Representation Account (DRA) of intrusive memories. In two pre-registered experiments using Bayesian sequential designs, we tested predictions derived from the DRA. In Experiment 1, we first aimed to replicate the functional dissociation between voluntary and involuntary memory after a visuospatial interference intervention, while matching retrieval conditions more closely than in previous research. Second, we tested the prediction that the effect of the intervention would be greater in an analogue trauma condition than a neutral control condition. Third, we tested the prediction that voluntary memory would be decreased in the trauma condition relative to the neutral condition. Finally, we hypothesized that measures of voluntary and involuntary memory would not be correlated across participants. Unexpectedly, the results indicated that the visuospatial intervention did not decrease involuntary memories and that measures of involuntary and voluntary memory were positively correlated. In Experiment 2, we focused specifically on replicating the effect of the visuospatial intervention on involuntary memories of a trauma film. We did not replicate this effect: results provided substantial Bayesian evidence against an effect of the intervention. Overall, our findings provide evidence against the DRA as a viable theoretical account of intrusive memories, at least in the context of a laboratory analogue of trauma.

Public Significance Statement

This research shows that a widely discussed theory about posttraumatic stress disorder, which provides an account as to why traumatic memories intrude into everyday life, may not accurately explain those experiences, at least when tested under controlled laboratory conditions with analogue trauma. The findings point to a need to rethink how intrusive memories work and how trauma may impact our ability to deliberately remember an experience. Moreover, they challenge previous evidence that a simple visuospatial task can reduce immediate lab-based measures of intrusive memories.

Keywords: intrusive memories, involuntary memory, trauma film paradigm, posttraumatic stress disorder

The dual representation account (Brewin et al., 2010; henceforth, DRA) for posttraumatic stress disorder proposes that during a traumatic event, intense emotional arousal up-modulates activity in a network comprising the amygdala, insula and early sensory cortical and subcortical areas, which strengthens the encoding of affective and sensory elements of the event. This explains why following the traumatic event, intrusive memories typically take the form of vivid mental images such as sights and sounds. At the same time, emotional arousal downregulates processing in hippocampal, parahippocampal and other structures in the medial temporal lobe which implement associative binding of the different elements of the episode (Figure 1). This results in ineffective integration of the affective and sensory elements with the spatiotemporal contextual details of the event, resulting in a fragmented episodic memory (Brewin et al., 2010). Together, these changes are thought to explain why sensory cues and feelings that are similar to aspects of the traumatic event can trigger unwanted involuntary memories of the event (henceforth: intrusive memories) even in spatiotemporally-dissimilar contexts. The core assumption of the DRA is that there are in effect two memory traces – sensory and contextualised representations – and that “pathological” encoding of trauma strengthens the former but weakens the latter, leading to dissociations between involuntary and voluntary memory.

The DRA was inspired by clinical findings that intrusive memories in posttraumatic stress disorder (PTSD, American Psychiatric Association 2013) are often fragmented (Brewin et al., 1996; for an overview see McNally, 2022), although the adequacy of this observation has been questioned (Rubin et al., 2008). One approach to studying intrusive memories in the laboratory uses emotionally distressing films as an experimental analogue of trauma exposure. In this “trauma film” paradigm, participants encode film clips that depict scenes with traumatic content, such as “actual

or threatened death, serious injury, or sexual violence”, following from the DSM-5 definition of trauma (American Psychiatric Association, 2013), under conditions that encourage personal involvement and immersion. Healthy participants typically report that segments of the films come to mind involuntarily for several days after viewing such films (e.g. Kobelt et al., 2024; Weidmann, Conradi, Gröger, Fehm & Fydrich, 2009; for review, see James et al., 2016). Same-day lab-based measures have also been used rather than just diary methods in real life (James et al., 2015; Lau-Zhu et al., 2019, 2021).

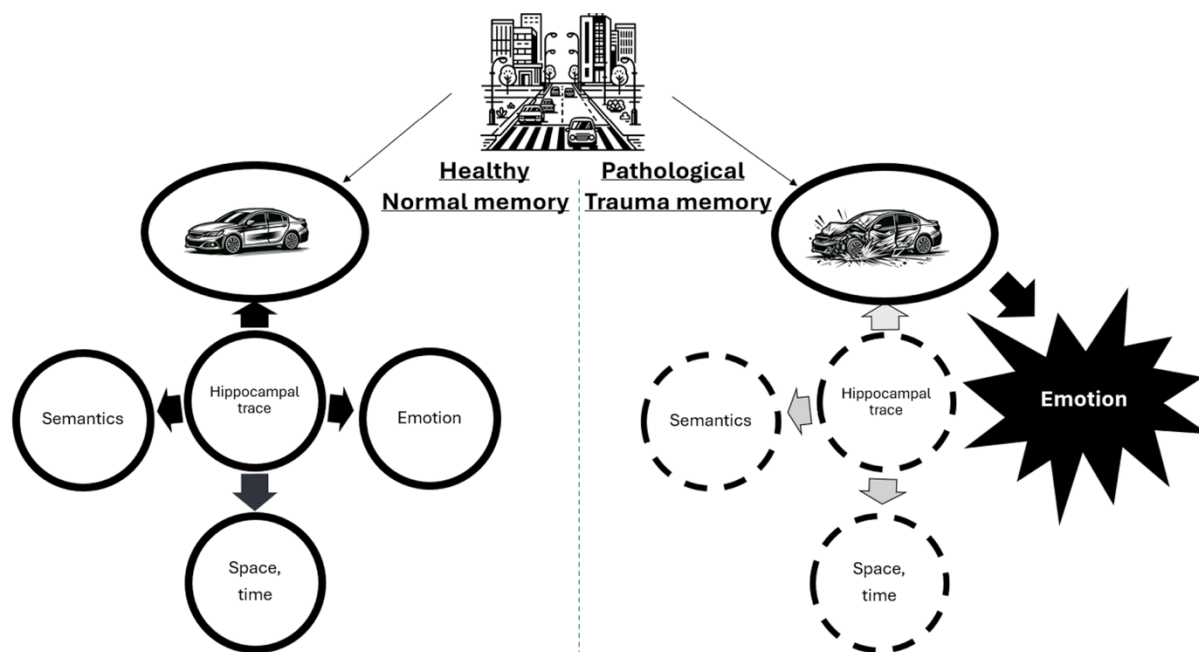
This evidence suggests that the trauma film paradigm can act as a laboratory model of exposure to trauma and subsequent intrusive memories of the experimental trauma, e.g. in PTSD (James et al., 2016). Brewin has recently emphasized that despite the difference between real trauma and lab analogues of trauma, the trauma film paradigm “has been very productive... in situations that demonstrate significant effects among participants all exposed to such a film. For example, it has provided evidence of a robust dissociation between the effects of a visuospatial intervention on voluntary and involuntary memory of traumatic scenes” (Brewin & Field, 2024, p. 2). This quote suggests that as long as the trauma film paradigm elicits intrusions in daily life, it provides a good model of pathological encoding, even in healthy participants. Indeed, it would be odd if there were two separate mechanisms in the brain that generate intrusive memories.

Crucially, the frequency of intrusions reported in the trauma film paradigm has been found to be unrelated to the accuracy of voluntary recall of the same material (Holmes et al., 2004), capturing the aforementioned characteristic of PTSD, i.e., that voluntary and involuntary recall can dissociate. Furthermore, certain visuospatial

interventions performed after film-viewing selectively decrease intrusive memory, but do not appear to decrease voluntary memory for the same films (Asselbergs et al., 2024; Deeptose et al., 2012; Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020; Lau-Zhu et al., 2019, 2021). The DRA has been used to interpret the results of the trauma film paradigm, especially findings of decreased involuntary memory for emotionally negative, arousing content following a visuospatial interference task (James et al., 2015; Lau-Zhu et al., 2019, 2021). The functional dissociation between voluntary and involuntary memory, which we refer to as the 'selective interference effect', provides particularly strong support for the DRA. For example, James et al. (2015) state that their "procedure modified intrusion frequency while leaving recognition memory intact, which indicates that trauma-film memory had not been erased but ceased intruding involuntarily. Although consistent with clinical models (Brewin, 2014), the dissociation between intrusions and recognition is contrary to traditional memory models (e.g., Tulving, 2002)—a paradox requiring future research."

Figure 1

Schematic of the Dual-Representation Account (DRA) of Intrusive Memory



Note. A visual reminder cue (street scene, top) may activate a hippocampal trace of a normal memory (left, a previous car journey), resulting in recall of a specific past episode, including sensory details (visual details of the car), semantic and emotional details (who we were with, the purpose of the trip, how much fun we had) as well as spatiotemporal context (where we were going and when). When encoding of trauma content is pathological (right, a car accident) it weakens the hippocampal trace of the trauma episode, symbolized here in dashed lines and lighter-shade arrows, while strengthening links between the hippocampal trace and the sensory details of the trauma. Thus, although the cue (street scene, top) triggers sensory and emotional experience (mental image of the crashed car, fear), it no longer activates the appropriate spatiotemporal contextual details of the event (when and where the car accident took place).

By positing two separate traces following pathological encoding of trauma, the core assumption of the DRA contrasts with a core mainstream assumption in memory research, which considers that all manifestations of episodic memory – both voluntary

and involuntary – originate from a single hippocampus index that binds together all of its elements, including its spatiotemporal context and the multimodal sensory representations (Frankland & Bontempi, 2005). We refer to this view as the ‘unitary’ account of memory. Specifically, the assumption of two separate traces in the DRA contrasts with unitary conceptualisations of PTSD that consider intrusive memory and voluntary memories to be based on the same memory trace (Berntsen & Rubin, 2008; Cohen & Kahana, 2022; Rubin et al., 2008). For example, a recent formal model of emotional memory, which was developed to explain recall of neutral words in laboratory experiments, has been applied to emotional memory and to intrusive memory in mental ill-health (Cohen & Kahana, 2022; Talmi et al., 2019). The model simulates each item or experience as a one-hot vector, the neural machinery that implements memory in the brain as a two-layer neural network and encoding and retrieval processes as operations carried out within this network. The emotional Context Maintenance and Retrieval (eCMR) and CMR3 models simulate emotional experiences through the value of the elements in the vectors that represent the experience. These models are supported by their ability to predict patterns of results in the human experimental and clinical psychology literature. Crucially for our purposes, CMR3 explicitly simulates the retrieval of experiences in the same way regardless of whether retrieval was intentional or involuntary.

Evidence that supports the unitary account of emotional memory includes the robust finding that voluntary memory of emotionally-arousing stimuli is enhanced when tested with methods that are known to tap on to hippocampal memory traces, such as free recall (Talmi, Luk, et al., 2007; Talmi, Schimmack, et al., 2007), cued recall (Dolcos et al., 2004) and recollection (Mihaylova et al., 2019; Rimmele et al., 2011, 2012; Yonelinas & Ritchey, 2015). The effects of emotion on voluntary memory are

169 attributed to emotional intensity rather than valence or the type of the emotion (Mather
170 et al., 2015). Additionally, in two studies of paired-associate cued recall, activation in
171 the hippocampus predicted recall of both negative and neutral stimuli, i.e., was not
172 modulated by the valence of the material (Bisby et al., 2016; Madan et al., 2017). The
173 findings that hippocampal-dependent memory for laboratory stimuli that are negative
174 and arousing is enhanced, rather than being impoverished, contradict the predictions
175 derived from the extension of DRA to milder emotional stimuli, and by so doing,
176 support the unitary account.

177 It is possible that the relatively mild manipulation of emotion in the
178 aforementioned studies was insufficient to weaken associative binding. However,
179 these studies fall within the scope of the extended explanatory scope of the DRA, as
180 it is currently used in emotional memory research. For example, in their joint review,
181 Bisby, Burgess and Brewin (2020, p. 270) show “how negative events can disrupt
182 these processes in healthy individuals and have argued that these mechanisms are
183 impaired in PTSD, resulting in impaired binding and holistic retrieval in this disorder.
184 Studies investigating such basic mechanisms provide a valuable addition to clinical
185 studies of trauma narratives and will... be crucial in providing further insight into
186 memory disruptions in PTSD”. That said, to test predictions derived from the DRA, it
187 may be necessary to employ stimuli that evoke stronger emotions, such as those used
188 in the trauma film paradigm.

189 Experiment 1 was conducted to test three key predictions derived from the DRA
190 using the trauma film paradigm. This paradigm was employed in a between-
191 participants design, with the factors of intentionality of retrieval (voluntary vs.
192 involuntary memory), emotion (traumatic vs. neutral film content) and intervention

(visuospatial interference vs. auditory control task). Prior to either intervention, we presented participants with reminder cues of the film, such that the reminder occurred in both intervention conditions. In the visuospatial interference intervention, the reminder was followed by the computer game of Tetris, which participants play under instructions to utilise mental rotation. This type of procedure has been shown to reduce intrusive memories in a number of previous studies (Asselbergs et al., 2024; Badawi et al., 2020; Deeproose et al., 2012; Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020; Lau-Zhu et al., 2019, 2021). However, it is important to note that other studies with possible variations on the intervention procedure have not observed a reduction in intrusions (Asselbergs et al., 2018; Brennen et al., 2021; Broughill et al., 2025; Brühl et al., 2019; Matura et al., 2025), while others show mixed effects, such as a reduction in shorter term but not in longer term intrusive memories (Wessel et al., 2025). A recent meta-analysis of laboratory studies shows that, on balance, this type of procedure decreased the frequency of intrusions (Varma et al., 2024).

Our main measure of voluntary and involuntary memory was the number of memory reports in a Vigilance Retrieval Task (VRT) designed to closely replicate the task employed by Lau-Zhu et al. (2019, 2021) on the same day as film viewing. After watching a film, participants in the VRT perform a primary digit-monitoring go-no go task, which demands most of their attention. Occasionally, blurred frames from the preceding target film and from an unseen foil film are presented behind the digits. This is supposed to mimic memory intrusions in everyday life, when one's attention is directed elsewhere, but a cue resembling a prior experience triggers the unintentional retrieval of a memory for that experience. The participants' secondary task is to press a key when such a memory comes to mind, at which point the primary task stops and they are asked to describe that memory. The main outcome variables are 1) the total

number of button presses (intrusions) and, by analysing the memory descriptions, 2) the number of unique film details reported, excluding repetitions of recurring details. As is typical with voluntary recall of films descriptions were coded according to the Autobiographical Interview (Levine et al., 2002). This is the first time such scoring is applied to analyse involuntary memory data after a trauma film. As is typical for trauma film studies, participants also completed diaries over the subsequent few days, recording intrusions from the films that occurred during their everyday lives, as an adjunctive measure of involuntary memory, albeit briefer to reduce burden, since it was not our main focus. The diary data were used as a manipulation check and supplementary measure for comparison with previous studies.

Experiment 1

The four hypotheses of Experiment 1 are shown in Table 1 and elaborated below.

Table 1

Experimental Hypotheses

Hypothesis (used for stopping?)	DRA-derived Hypothesis	Fixed factor (level)	IV1 (between)	IV2 (between)	DV
A (yes)	Intentionality of retrieval X Interference Visuospatial interference decreases involuntary memory for films with traumatic content more	Emotion (Negative film)	Intentionali ty of retrieval (Voluntary, Involuntary)	Interference (Visuospatial, control)	1,2

	than voluntary memory.				
B (yes)	Emotion X Interference Visuospatial Interference decreases involuntary memory more in the trauma than the neutral condition	Intentionality (Involuntary)	Emotion (Negative, Neutral)	Interference (Visuospatial, Control)	1,2
C	Conditional on the presence of involuntary trauma memory: Traumatic film content decreases voluntary memory	Interference (Control) Intentionality (Voluntary)	Emotion (Negative, Neutral)		2
D	Voluntary and Involuntary memory will not be correlated across participants	Intentionality (Involuntary first)			1,2

235 Note. DV1 refers to the number of memories reported in a given test, DV2 refers to
 236 the number of unique details within reported memories (see methods sections for
 237 detailed definitions).

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 240
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Hypothesis A: Visuospatial interference decreases involuntary memory for traumatic film content more than voluntary memory

This hypothesis is based on a core claim of the DRA: “The extent of voluntary recall and the number of involuntary images of the same traumatic materials appear to be functionally independent” (Brewin, 2014, p. 85). In this section, we query which dependent variable operationalises this claim most appropriately. The literature on involuntary autobiographical memories distinguishes between the number of unique memories that come to mind and the frequency with which they do so (Berntsen & Rubin, 2008). Similarly, in the trauma film paradigm, one can measure the number of unique intrusive memories (e.g., from different scenes in the film), or their total number, including those with repeated content. For example, if a participant has three intrusions in total, two that are the same and relate to event A and another that relates to event B, they will be scored as having three intrusive memories (henceforth: DV1) but only two unique ones (henceforth: DV2). Typically, only a small proportion of total intrusions reflect unique memories. This is because only a few ‘hotspots’ in the films are perceived as the “worst moments” for any one individual, but memories of the same hotspots may come to mind many times. The measure of total intrusions has greater clinical significance as an analogue of the extent of intrusive memory causing disruption in PTSD and therefore often forms the key dependent variable in the trauma film paradigm. This is why, in most previous studies that used diaries to record intrusions, participants were instructed to record every intrusion in their week-long diary, even if from the same memory. By contrast, the typical measure in voluntary memory tests is a count of unique studied items that participants can recall or recognise among foils. This raises the possibility that the differential effect of the

intervention could be due to using different measures in the involuntary and voluntary conditions.

To appreciate the significance of this observation, consider that the voluntary memory measures in most previous studies of the trauma film paradigm were cued recall or recognition tests, where it is impossible for participants to repeatedly retrieve the same stimuli. Repetitions are, in principle, possible in free recall tests of memory, but are often not reported. In practice, free recall tests encourage “retrieval monitoring”, defined as the search and decision processes operating during recollection attempts (Koriat & Goldsmith, 1996). Consequently, participants do not typically repeat what they have already said (Zaromb et al., 2006).

In addition to distinguishing two types of memory performance (DV1 and DV2), we changed three aspects of the previous free recall protocol of Lau-Zhu and colleagues (2019) to better match the voluntary and involuntary measures. First, both voluntary and involuntary memory measures were collected under the matched conditions of the VRT. While previous studies tested involuntary memory using free report (of intrusions), most of these studies tested voluntary memory using recognition (e.g. Holmes et al., 2009), a measure that does not necessarily entail hippocampal-dependent associative retrieval (e.g., Yonelinas et al., 2001). In the only study to test both voluntary and involuntary memory using free report, voluntary memory was tested under full attention conditions, but involuntary memory was tested under the dual-task conditions of the VRT (Lau-Zhu et al., 2019). Second, our instructions encouraged participants not to censor repetitions in either involuntary or voluntary conditions. This was helped by the fact that memories were reported by typing into a text box, which disappeared when they had finished, so participants could not refer back to what they already retrieved (cf. previous studies using written recall on paper). Third, we

manipulated retrieval intentionality between-participants. The two studies that reported the key dissociation between voluntary and involuntary memory using the VRT (Lau-Zhu et al., 2019, 2021) used a within-participant design, where the involuntary test was given first, followed by the voluntary test. This not only results in unequal retention intervals, but also output interference, where the reports from the first test interfere with reports from the second test. A within-participant design may also encourage adoption of various strategies in the second test, such as explicit attempts to recall items reported in the first test or to suppress such items.

To summarise, we extended the previous studies cited above that reported a dissociation between the effect of visuospatial intervention on voluntary and involuntary memory by making important methodological improvements to a laboratory measure of intrusions following the trauma film paradigm. Using the VRT, we scored both the total number and number of unique memories, equated the memory cues and level of attention, matched the retrieval instructions, and eliminated test order effects. The functional dissociation reported in previous studies could have arisen from one or more of these methodological differences between voluntary and involuntary conditions. We aimed to test a key prediction derived from the DRA, that a visuospatial intervention decreases involuntary but not voluntary memory on the VRT. While replicating the previous functional dissociation would bolster support for the DRA, a lack of such an interaction would favour unitary accounts of voluntary and involuntary memory.

Hypothesis B: Visuospatial Interference decreases involuntary memory more in the traumatic than the neutral content condition

In his seminal 2014 paper, Brewin attributes the functional dissociation between the effect of a visuospatial intervention (vs. the control intervention) on involuntary vs. voluntary memory for traumatic experiences to dysfunctional encoding during trauma that is unlikely during neutral experiences. This claim is central to the present paper, so it is worthwhile to quote in full: “Engaging in visuospatial tasks at or shortly after trauma encoding appears to block the development of involuntary images, whereas engaging in a verbal task that specifically involves counting backward leads to more involuntary images; the extent of voluntary recall and the number of involuntary images of the same traumatic materials appear to be functionally independent... These data are very largely based on viewing of trauma films, and the effects may not be observed with other stimuli... stimuli that have no specific personal meaning may make it unlikely that dissociation, a typical response to traumatic situations, will occur” (Brewin, 2014, p. 85). This statement clearly implies that predictions derived from the DRA entail that we should only observe a significant interaction between intentionality and the intervention in the trauma condition, but not in the neutral condition.

To quote again from Brewin, “visuospatial tasks are particularly likely to interfere with perceptual, image-based representations” (Brewin, 2014, p. 80). The DRA considers involuntary memories of trauma, either in PTSD or in the trauma film paradigm, to be image-based. Supporting the possibility that they are particularly image-based, one previous laboratory study has compared the phenomenology of involuntary memories for trauma and neutral content, and found that involuntary memory of the trauma content was more image-based than verbal-based (Hagenaars et al., 2010).

Involuntary memories also follow non-trauma experiences and are common in our daily lives (Berntsen, 2021). Outside the lab, negative involuntary memories are

prevalent in many mental health conditions, where they are often associated with a range of stressful events (e.g. a moment of being deeply shamed by peers at school associated with depression), not just DSM-5 trauma, and positive involuntary memories have been reported in, for example, hypomania and food cravings (Brewin et al., 2010; Holmes & Mathews, 2010; Malik et al., 2014). In the lab, involuntary memories have also been documented following positive and depressive film material (Clark et al., 2013; Davies et al., 2012; Lang et al., 2009). Involuntary memories for non-trauma content can also be image-based. This possibility is supported by findings that involuntary memories were rated as clearer and more vivid than voluntary memories, and that some of them were rated as more relevant to the task situation in which they came to mind (Barzykowski & Staugaard, 2016). To the extent that involuntary memories in our neutral condition are image-based, then based on the DRA, the visuospatial intervention could decrease them.

Taken together, these two quotes lead to apparently contradictory inferences about what the DRA would predict for our neutral condition. The first (trauma-based) suggests that a visuospatial intervention would decrease involuntary memory only in the trauma condition, while the second (image-based) suggests that it would decrease neutral involuntary memory as well, consistent with the findings of Davies et al (2012) for involuntary memory after a positive film.

The DRA has not considered neutral involuntary memories much, and this may be the reason for the apparent contradiction. This is because according to the DRA, there would be more involuntary memory following a traumatic than a neutral experience. A core assumption of the DRA is that pathological encoding of a trauma weakens hippocampal-dependent contextual representations, so that these representations do not inhibit the automatic, unwanted retrieval of sensory

representations that are routinely cued by the environment. By contrast, memories of less emotional experiences are less likely to be encoded in a pathological way and are, hence, integrated with their spatiotemporal context and less likely to come to mind. If the number of involuntary memories in the trauma condition is very different than in the neutral condition in our control condition, and visuospatial intervention has the same proportional impact in both conditions, it would decrease the *number* of voluntary memories more in the trauma condition. Taken together, we argue that based on the DRA, an interaction would be expected between the factors of intervention and emotion, owing to proportional effects of the intervention applied to different baselines.

To test this prediction and illuminate the debate, we adapted the trauma film paradigm for viewing neutral films without traumatic content. We opted to use films that most participants experience as neutral because in most past studies, the non-trauma films were still emotionally evocative, and therefore, it is possible that strong arousal up-modulated their sensory representations (Mather et al., 2015); even if this effect has been particularly apparent for negative content (Bowen et al., 2018). We administered the neutral and the traumatic films to separate groups of participants, half of whom received the visuospatial intervention and half the control intervention. Our predictions derived from the DRA are that there will be more involuntary memories following the trauma condition compared to the neutral condition, and that emotion and intervention would interact, such that the visuospatial intervention will decrease the number of involuntary memories more in the trauma condition.

Finding that a visuospatial intervention decreases involuntary memories to the same extent regardless of their emotionality would challenge the DRA by decreasing the case for evoking separate representations for trauma situations specifically, given

the DRA was at heart derived to account for the pathological encoding of trauma. Such a finding would instead be consistent with a unitary account of memory. It would suggest that the functional dissociation between the effect of the visuospatial intervention on voluntary and involuntary memory reported in previous research is not due to quasi-pathological encoding of trauma but applies just as well to healthy encoding of neutral content. Furthermore, it would suggest that there may not be mechanistic differences between involuntary memory in everyday life and intrusive memory in PTSD, as assumed by some researchers. As such, it would help evolve the unitary account of memory by encouraging research on how a single trace can support both voluntary and involuntary memories, both neutral and traumatic. Likewise, it might also help evolve future versions of DRA, which has not yet been extended to account for neutral memory.

Hypothesis C: Trauma decreases voluntary memory

Because intact hippocampal function is essential for the ability to freely recall previously-studied verbal and visual stimuli (Isaac & Mayes, 1999a, 1999b), and a core assumption of the DRA is that pathological encoding of trauma weakens hippocampal-dependent contextual representations, it follows that based on the DRA, pathological encoding of trauma would impair voluntary free recall, while healthy encoding of trauma or of non-trauma content would enhance voluntary recall. This is clear, for example, in this quote: “According to dual representation theory, perception of a moderately stressful or emotionally salient event results in the creation of more enduring C-reps and S-reps.” (Brewin et al., 2010, p.221). The balance between healthy and quasi-pathological encoding in the sample allocated to the voluntary

trauma condition will determine whether average recall performance in that condition is lower or higher than performance in the voluntary neutral condition.

In their 2020 review, Bisby, Brewin and Burgess discussed experimental paradigms that contrast voluntary memory for negative and neutral content and argued that lower associative memory performance in the negative condition reflects mechanisms that are impaired in PTSD. This suggests that healthy samples tend to encode even milder emotional situations in a quasi-pathological way, measurable in indices of associative memory. Based on this extension of the DRA to milder emotional situations we can derive a more clear-cut prediction: that recall of trauma films will be decreased, compared to films with neutral content. This interpretative framework is gaining dominance in emotional memory research (e.g., Clewett & McClay, 2024).

The robust finding of enhanced free recall of emotional material appears to challenge this extension of the DRA. It has been robustly demonstrated that when people view emotional and neutral stimuli, such as words and images, they typically recall more of the emotional than neutral stimuli (Ack-Baraly et al., 2016; Talmi, 2013). However, as mentioned before, any effects in these studies may be due to the relatively mild emotional arousal they induced in participants. The trauma film paradigm employs specific instructions to increase engagement with the film content to enhance its utility as an analogue of exposure to trauma. Brewin and Field (2024) suggest that the presence of intrusive memories in setups that resemble the current involuntary trauma condition is indicative that some pathological encoding has occurred.

The prediction derived from the extension of the DRA therefore suggests that voluntary recall of trauma content will be decreased, compared to recall of neutral content, because free recall relies on associative processes, and these should be

weaker when content is emotionally arousing and negative. Failure to observe this pattern will help develop the DRA by clarifying how it applies to milder emotional manipulations.

All our conclusions will be inevitably tempered by the degree to which our outcome variables reflect each type of memory representation accurately and reliably, but this limitation is particularly acute for this latter prediction where the degree of pathological encoding may depend on stimulus and individual differences. Given the nuance involved in developing Hypothesis C, we treat it as having secondary importance compared to Hypotheses A and B. We therefore planned to only terminate our sequential design based on our primary Hypotheses A and B (see Methods).

Hypothesis D: Involuntary memory will not predict subsequent voluntary memory of the trauma film

Examining both voluntary and involuntary within the same participants allows us to test another prediction of the DRA. The DRA contends that there is a “functional independence between perceptual and episodic memory systems” (Brewin, 2014, p.80). This contention is supported by evidence that voluntary and involuntary memory are not correlated across participants (e.g., Tables 3 and 4 in Brewin, 2014). Our study provided an opportunity to examine this question afresh, with the benefit that the voluntary measures of memory is more closely matched to the involuntary measure than in previous studies, and that we extract two dependent measures, typical to each research literature. Thus, our fourth hypothesis (Hypothesis D, Table 1) was that voluntary and involuntary memory would not be correlated across participants. Since Brewin criticised studies that employed a voluntary test prior to the involuntary test, and that administering a voluntary memory test for a trauma film has been shown to

decrease subsequent involuntary memory reports (Krans et al., 2009), we only examined this correlation on the half of the sample who undertook the involuntary test first.

Methods

This in-person laboratory study used a 2x2x2 between-participants factorial design, with factors of intentionality (voluntary vs. involuntary memory), emotion (films with trauma vs. neutral content) and intervention (visuospatial interference vs. control task). Participants were randomly allocated to one of the eight conditions.

To compare involuntary (intrusion) and voluntary memories of film content most effectively, the task used should be tightly matched between both conditions. A tight match ensures that the previously reported dissociation between intrusions and voluntary memories cannot be traced back to differences in task parameters. Previous dissociations between intrusive and voluntary memory have been based on very distinct tests, such as a diary (intrusion; mostly taken over 7 days in the participants' daily life) versus lab-based tests for voluntary memory (recognition memory, which involves a lower element of search; or free recall). The Vigilance Retrieval Task (VRT), developed by Lau-Zhu and colleagues to measure intrusive memory (Lau-Zhu et al., 2019, 2021) and here extended to also measure voluntary memory, permits closer matching of these parameters. This is because the VRT can be administered in the same environment (lab), with the same number and timings of memory cues (blurred images from the films distributed throughout the task), involves the same primary task (number counting) and output format of memories (typed). We thus used the VRT as a primary outcome measure for both involuntary and voluntary memories in a

between-subject design (to keep the retention interval the same). Following this part of the experiment, participants undertook a second VRT, as described below.

We also included a diary measure, for the purpose of allowing a comparison to previous studies of intrusive memories (since diary is the usual primary outcome in such studies). The intrusive memory diary was conducted online rather than on paper, as is increasingly common in research in this area (Espinosa et al., 2023; Singh et al., 2023). Given our focus on the VRT, the total duration of the lab-based session, and the high recruitment target, we opted for a briefer version of the diary with minimal demands on participants (on the day of the experimental session and the following 3 instead of 7 days), and single rather than multiple prompts per day (1 time slot per day rather than the usual 3). We chose this time-period because intrusions tend to be highest in the first three days following the intervention (James et al., 2015) and to only have one time slot to reduce participant burden.

It is important to manipulate intentionality of retrieval between-participants, because otherwise, and as discussed above, test order effects, such as those due to output interference, may confound the results. However, it is also advantageous to measure voluntary and involuntary memory within the same participants, because the specific hotspots that an individual experiences as “worst moments”, and therefore later become intrusive memories, vary across individuals (Holmes et al., 2005; Grey & Holmes, 2008). Thus, in an experimental context one way to tell whether a particular event was a hotspot for someone is to check whether it comes to mind in an involuntary memory test. It follows that we can only examine voluntary memory for hotspots if we collect both voluntary and involuntary measures in the same participant. Although previous comparisons between voluntary and involuntary memory measured each type of memory separately, rather than conditionalize one measure on the other, it is

theoretically important to do so because the voluntary/involuntary dissociation predicted by DRA specifically concerns memory for the hotspots themselves. In particular, the DRA prediction that a visuospatial intervention affects involuntary but not voluntary memories (Hypothesis B) is arguably restricted to hotspots within the film (since non-traumatic, non-hotspot moments should not intrude). It is possible that previous evidence of this dissociation may be due to voluntary recall measures being dominated by details from non-hotspots, which were not affected by the intervention. It is therefore possible that if we restrict the memory measure to hotspots, the intervention may affect involuntary and voluntary recall equivalently. Such a finding will challenge an important support for the DRA. These considerations led us to ask all participants to undertake two tests of memory in the VRT, a voluntary and an involuntary test, one after the other (their order counter-balanced across participants). The key Hypotheses A-C (Table 1) were tested only with data from the first VRT, so are immune to the effects of test order, but we also analysed the second test for Hypothesis D.

The trauma film paradigm has been argued to be a “productive” laboratory analogue of trauma (Brewin & Field, 2024), but changing the protocol could weaken this support. For example, Brewin and Field (p. 2) argue that “When findings are null... the paradigm is vulnerable to the criticism that this may be because it does not adequately model exposure to a traumatic event.” For this reason, we kept three features of the typical protocol. First, we manipulated the intervention task between-participants to avoid potential order effects in terms of the impact of the trauma or demand characteristics (Kehyayan et al., 2024). Second, we manipulated film content between-participants to keep the nature of the experience as similar as possible to the original (and use the same trauma films as those used in Lau-Zhu et al., 2019, 2021).

Finally, we collected diary measures of involuntary memory to strengthen evidence that we have adequately modelled exposure to a traumatic event.

Transparency and openness

As discussed transparently in the discussion section of this study, Experiment 1 was accepted as stage 1 registered report by PCI, available at this link (<https://osf.io/ur49s>). Due to a technical error, data collection stopped prematurely. Consequently, the manuscript was withdrawn from PCI (<https://rr.peercommunityin.org/articles/rec?id=807>). The general introduction, the introduction of Experiment 1, and the methods section are similar to the text approved by PCI, other than where textual changes were introduced to improve clarity and fit to the journal, and where explicitly noted.

We pre-registered computing DV3, voluntary memory for one's own hotspots, defined as the number of unique film details that are reported in both VRTs, but did not conduct analyses of this metric. We also did not conduct some secondary analyses that were proposed in the Stage 1 registered report. The experiment was also pre-registered here: <https://osf.io/fxj9n/overview>. Any deviations from pre-registered methods and results are explicitly stated. All data and analysis code are available on OSF (https://osf.io/cms7n/?view_only=0fa9878af33b40de9a54581c71af5217). Analyses that were pre-registered on the OSF as 'exploratory' are described as 'secondary analyses' below. Any non-registered analysis is reported as 'exploratory'.

For both this experiment and Experiment 2, we report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. The study follows Journal Article Reporting Standards (JARS) for Quantitative Research in Psychology (Appelbaum et al., 2018). Data were analysed using R,

version 4.3.2 (R Core Team, <http://www.r-project.org>) and the package *ggplot2* version 4.0.1.

To determine the sample size in Experiment 1 we used a sequential Bayesian design (Schönbrodt & Wagenmakers, 2018), planning to stop when the Bayes Factor (BF) was greater than 6 for either the alternative (BF_{10}) or null (BF_{01}) model for each of our two main hypotheses (A and B), using DV2. To minimise “false positives”, we planned to collect initial batches of $n=30$ in each of the 8 groups, followed by batches of $n=10$ per group if our BF criteria were not met, up to a maximum of $n=60$ per group (i.e., $N=480$ participants in total) for practical (resource) reasons. Previous trauma film paradigm studies have estimated a main effect size of the intervention (compared to control) on involuntary memory with Cohen’s $d=.91$ (Lau-Zhu et al., 2019; based on diary data from Holmes et al., 2009). The effect of the intervention on the VRT in Lau-Zhu et al. (2019) was 0.96 and 1.20 in Experiments 2 and 3, respectively. Given that neither interactions of the intervention with emotionality of the film nor intentionality in the VRT have been tested thus far, we simulated interaction effects of approximately half this size, i.e. Cohen’s d of 0.5 ($f^2=0.06$). If both hypotheses A and B have this effect size, simulations with the *anovaBF* function showed 78% probability of finding $BF_{10}>6$ for both (with median of 50 per group after stopping), or >99.9% probability of finding such evidence for at least one of these hypotheses. If both of the two null hypotheses were true, then at the maximum N , the “false positive” rate was importantly low at 1.3%. However, in a deviation from our pre-registration, a coding error meant that the script instructed us to stop when $BF_{01}>3$ rather than $BF_{01}>6$. With this more liberal threshold, we had a 55% chance of supporting both nulls, or 89% chance of supporting at least one null.

To minimise any potential bias during data analysis, at each stopping point, an automated analysis script was used, which provided only either a 'stop' or 'go' answer but not the BFs associated with each hypothesis (Sadibolova & Terhune, 2025).

Participants

At the point of stopping (see Transparency and Openness section above), the sample size was 240 (30 per group), composed of 145 females, 94 males and one non-binary individual. The average age was 29.20 ($SD = 11.18$). Participants were recruited from the local community through the Cambridge SONA platform. Participants were healthy volunteers, between 18-68 years old, fluent in English, who were able to comply with the study protocol. They were screened for conditions that may influence brain function (a current mental health diagnosis, currently treated for substance abuse, a neurological disorder, history of head injury, taking medication or drugs that influence brain function). Participants who were undergoing particularly stressful life events were advised not to take part. The Cambridge Psychology Research Ethics Committee has given favourable opinion to this study (reference No: PRE.2024.010).

Materials

Each film consisted of 11 audiovisual clips sourced from the public domain and had a total duration of 12 minutes. The trauma film was the same as in our previous studies (Lau-Zhu et al., 2019, 2021) and featured content showing types of events listed as traumatic events in the DSM-5, including road safety films featuring car accidents, scenes of war, and surgical close ups. The films selected for the neutral condition included non-threatening content, such as cooking, driving and

using an exercise machine. The films were presented on a wall projector in a darkened room, with sound delivered via headphones.

In previous studies employing neutral and negative film paradigms, some authors selected films that differed primarily in emotional tone without systematically matching other stimulus characteristics (e.g., Chouinard-Gaouette & Blanchette, 2025; Hørlyck et al., 2019), while others additionally controlled thematic content (e.g., traffic vs. food scenes) and film length (Arnaudova & Hageraars, 2017; Ferree & Cahill, 2009; Hageraars et al., 2010; Kobelt et al., 2024). More superficial visual attributes are more rarely controlled (Arnaudova & Hageraars, 2017; Kobelt et al., 2024), as are narrative qualities (Ferree & Cahill, 2009, Taylor et al., 2021; 2025). Here, we matched the number and duration of clips in the negative and neutral films and included people in every clip. Instead of content, we matched the semantic relatedness of the clips in the trauma and neutral condition, a variable with known influence on memory recall (Talmi & Moscovitch, 2004).

The emotional and neutral films were matched based on ratings solicited from online pilot participants ($n = 19$; 10 females, 9 males; age $M = 33.25$, $SD = 11.97$). As expected, emotional clips were rated as more negative than neutral clips ($M = 6.99$, $SD = 0.68$ vs. $M = 4.23$, $SD = 0.64$), with a Bayesian t-test providing extreme evidence for this difference, $BF_{10} = 6.19 \times 10^6$. Emotional clips were also significantly more arousing ($M = 5.66$, $SD = 1.45$) than neutral clips ($M = 2.56$, $SD = 1.10$), with a Bayesian t-test providing extreme evidence for this difference, $BF_{10} = 4.40 \times 10^4$. Emotional clips were rated as having a significantly higher subjective impact ($M = 5.68$, $SD = 1.59$) than neutral clips ($M = 2.40$, $SD = 1.05$), with a Bayesian t-test providing extreme evidence for this difference, $BF_{10} = 2.20 \times 10^6$. These results supported the differentiation of the two stimulus sets. Finally, pairs of film clips were

rated for similarity on a 1-7 scale. Inter-rater reliability of similarity judgments was assessed using an intraclass correlation coefficient (ICC). Reliability was good for neutral clip pairs (ICC = 0.80) and excellent for emotional clip pairs (ICC = 0.91), indicating strong agreement across participants. The final selected 11 clips from each Emotion condition had average similarity scores of 3.97 in the Negative and 3.94 in the Neutral condition, with a paired Bayesian t-test providing moderate evidence against a difference in similarity ($BF_{01} = 4.17$).

The reminder cues consisted of 11 film stills. One still was taken from each of the discrete clips in the film and presented in the same fixed order as the scenes within the film. The stills for the traumatic content condition are equivalent to Lau-Zhu et al. (2019), and correspond to frames just before the worst moments, which have been clinically associated with intrusive memories (Ehlers, Hackmann, & Michael, 2004). Examples are: a picture of a circus (prior to the elephant escaping and going on a rampage), or a smiling teenager (just before getting hit by a van while being distracted by texting). For the neutral content condition, stills were taken from the beginning of the film. Examples are: a picture of two women standing in a gym (before starting a tutorial about a leg-press machine), or a teacher in front of a classroom (where she goes on to teach her students about coordinate systems).

The images used in the VRT included blurred versions of 11 film stills (1 per movie, featuring similar content to the reminder cues; each displayed twice) and 68 foil stills (novel images featuring a variety of coloured indoor/outdoor scenes; each displayed once), shown in a fixed randomized order. The images were passed through a Gaussian filter using the MATLAB function *imgaussfilt* (sigma = 38), to approximate cues glimpsed in everyday life when outside one's focus of attention (Berntsen, 2009). The reminder cues and VRT were implemented using MATLAB

R2023b (Mathworks Inc., Sherborn, MA) running Psychtoolbox version 3 (Brainard, 1997; Pelli, 1997; Kleiner et al., 2007). The aspect of the VRT “please describe your memory here” (Figure 2) involved participants typing a description of their memory into an unconstrained text entry field as soon as they made each response (see Appendix for instructions both voluntary and involuntary VRT).

The visuospatial intervention condition involved playing the game Tetris (The Tetris Company, 2007). The game’s objective is to arrange different geometric objects into complete horizontal lines. The objects are of different shape and colour, can be rotated using the arrow key, and fall from the top of the screen. Whenever a full horizontal line is formed, it disappears, and the participant is awarded points. The game adapts to participants’ performance and is over when the screen has vertically filled up with the blocks.

For the control condition, we selected a podcast listening task, where after the reminder cue participants listened to a podcast (the start of an episode of the BBC The Public Philosopher, <https://www.bbc.co.uk/programmes/b043wx2s>) about voting in elections and answered simple content questions. Critically, listening to a podcast uses a different sensory modality than the largely visuospatial nature of Tetris. This is important since the DRA emphasizes the visuospatial aspect (Brewin, 2014), which means that any visual control task could be problematic. Additional advantages of the podcast task were that, like the visuospatial intervention, it also employed a digital activity from the public domain, and controlled for the time spent, nature of contact with researcher, instructions for active engagement; therefore providing more experimental control than passive rest. A translational advantage is that the podcast task has real-world clinical relevance, since it was used (with different podcasts) as a control when an imagery-based experimental intervention was delivered soon after

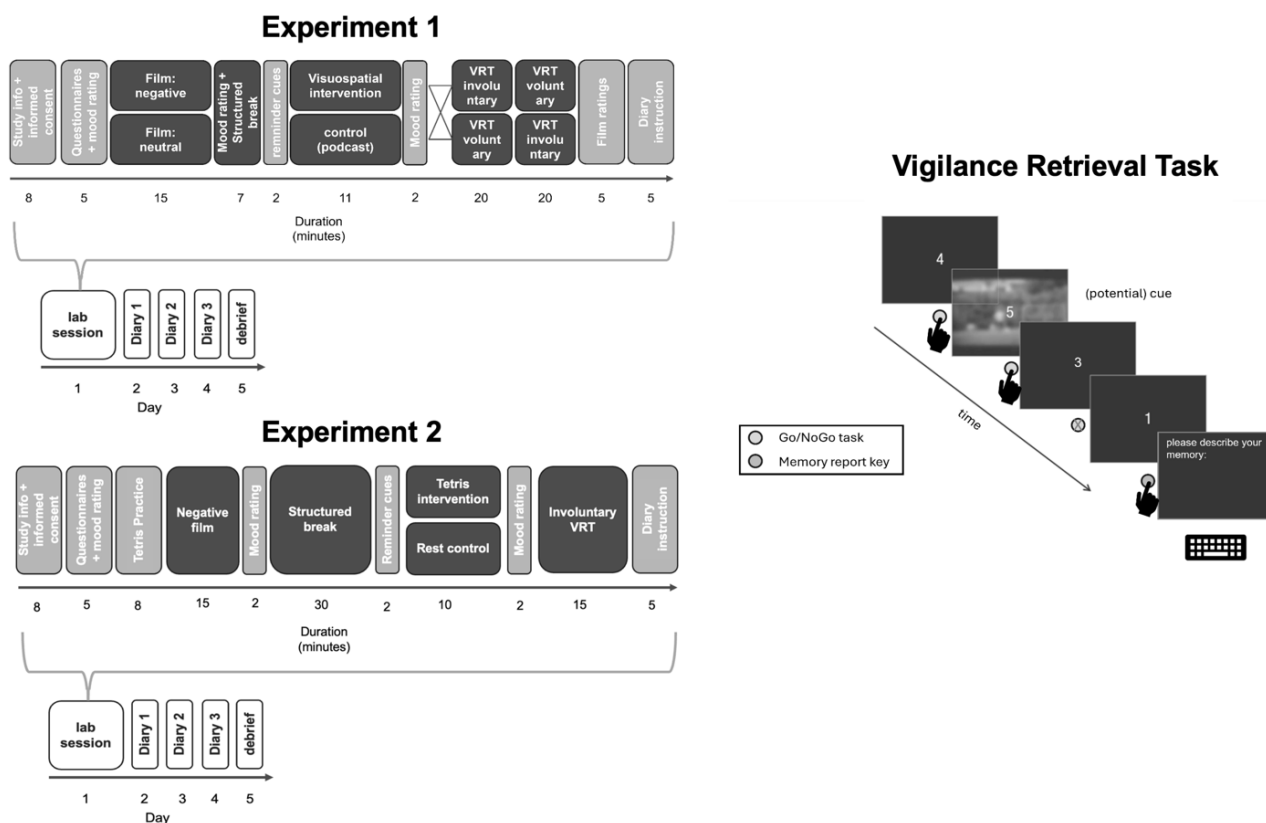
trauma in a hospital emergency department (Kanstrup et al., 2021) and for established intrusive memories (Kanstrup et al., 2024). The limitation of the podcast task is that the podcast and the visuospatial interventions have not been formally matched for cognitive load. Therefore, a limitation of this experiment is that the control intervention controls for some, but not all, aspects of the visuospatial intervention. Additionally, the effects of verbal interventions are mixed in the literature; a verbal pub quiz task (Holmes et al., 2010) increased intrusions, but a word game task (Hagenaars et al., 2017) decreased intrusions, relative to rest.

Procedure

Figure 2 depicts the timeline of the lab-based session.

Figure 2

Experimental Timelines and Vigilance Retrieval Task Layout



Note. VRT refers to the Vigilance Retrieval Task. All participants completed both versions of this task (voluntary and involuntary), with the order of the two tests manipulated between-participants. For the tasks marked in darker shade in the timeline, the experimenter left the room after instructions were given. For those marked in lighter shade, the experimenter remained in the room. In the VRT, participants pressed the 'Go' key for all digits displayed except for 3 (with the cross on the Go/NoGo task button indicative of a NoGo trial in which 3 is displayed). Cue images were shown in the background every three digits. Cue images were blurred stills, either from the film viewed or from previously unseen foil images. After each button press, memory descriptions were typed into an unconstrained text entry field (see Appendix).

We devised a standardised script as well as some protocol training and pilot cases for all experimenters to limit differences in delivery of the protocol across research participants. After receiving information about the study and providing their written and informed consent, participants completed the Spontaneous Use of Imagery Scale (Kosslyn et al., 1998) and a baseline mood rating questionnaire. Prior to film viewing, participants were instructed to pay full attention throughout the film, and to immerse themselves in it. The experimenter left the room during film viewing (trauma or neutral version; 12 min) but monitored participant's compliance with film viewing instructions remotely using a live camera feed of the testing room. After the film, participants completed another mood rating and questions about their attention to the film; and had a structured break in which they completed simple questions using a thesaurus (5 min). They then viewed brief reminder cues of the film in the form of still images from the film and completed either the visuospatial intervention or the control

task (10 min), followed by a post-intervention mood rating questionnaire. Next, they undertook two versions of the VRT, one after the other (approx. 20 min each).

The VRT task layout was the same in both voluntary and involuntary conditions, and the instructions were as closely matched as possible (see Appendix). At the end of the lab-based session, participants rated the eleven film clips they had seen on various attributes (see below). Finally, participants received instructions on how to fill in the diary questionnaires they would receive in the evening that day and over the following three days. On the day after completing the last diary questionnaire, participants were debriefed via email and received confirmation of reimbursement. Overall, the procedure and experimental materials are based on Experiment 2 of Lau-Zhu et al., 2019. The similarities and differences of the present study to Lau-Zhu et al. (2019) are listed in Supplementary Table 1.

Mood rating

Participants were asked to indicate, on a visual analogue scale ranging from 'not at all' to 'extremely', how sad, happy, anxious, depressed, calm, hopeless, or angry they feel. Mood ratings were taken at three time points; prior to watching the film, immediately after watching the film, and after completing the intervention.

Post-film questionnaire

Participants answered a series of questions about their film viewing experience using 0-100 scales, indicating how much attention they paid to the film; how immersed they felt in the film; the percentage of time they looked away from the screen and how distressing they found the film. In a deviation from our pre-registration, due to technical error, we did not ask participants how personally relevant they found the film.

Reminder cues

Eleven film stills were presented on the testing computer. One still was presented at a time (duration 3s per still), against a black background. Participants received the instruction to sit still and pay close attention to the images. To maintain experimenter blinding to Emotion condition, the experimenter faced away from the monitor during the presentation of the reminder cues. The lights were switched off during this task, to parallel the context of film viewing.

Intervention

Most previous studies of the trauma film paradigm have not blinded the experimenter to any of the manipulations. Here the experimenter was blinded to the allocation of the participant to the Emotion condition, which could otherwise have led to different demeanour. When we designed this experiment we believed that given that the experimenter needed to be in the room and give instructions to ensure and check participant comprehension and adherence to the protocol associated with the intervention, they cannot be blinded to the actual intervention. To mitigate demand characteristics, the experimenter only learned the intervention condition after the post-film structured break, before the reminder cues were shown.

Participants in the visuospatial intervention first received instructions on how to play the game, with the emphasis that they should mentally rotate the pieces to foresee where they will fall, and to pay attention to the three upcoming objects displayed at the right side of the screen. Both switching off the ghost piece and the mental rotation instructions were used to encourage participants' active use of mental rotation in the game. Instructions were followed by 1 minute of practice, after which Tetris was played in marathon mode (with 15 levels) for 10mins, with the sound off.

Participants in the control condition received instructions on how to listen to the podcast, with the emphasis to focus on the ideas and the meaning. They then received the question sheet and went through it with the experimenter. Participants listened to 1min of the podcast for practice, and then continued listening to the episode for a further 10 min.

Vigilance Retrieval Task (VRT)

We used the VRT to assess voluntary and involuntary memory in participants, with the instructions presented in Appendix. In previous studies, we have validated the VRT by showing that the number of intrusive memories in the task was associated with the number of intrusions in a 7-day diary and verified that brief reports provided during the VRT described actual film content (Lau-Zhu et al., 2019, 2021).

Participants completed a sustained attention Go/NoGo task, in which a stream of digits (1-9) was presented at the centre of the screen (250ms; 270 total trials). Participants were asked to respond with a button press to all digits, except for when a '3' was shown (11% of trials). The digits were either shown on a black background (67% of trials) or a blurred image (33% of trials). While participants were instructed to focus on the Go/NoGo aspect of the task, they were also asked to react with a separate button press whenever they remembered any of the clips from the previous film. In the involuntary condition, participants were asked to pause the task (by pressing a key) whenever a "mental image [of the film] pops into your mind spontaneously". The voluntary condition was matched to the involuntary condition, except participants were asked to deliberately recall film content and pause the task whenever they succeeded (meeting the "retrieval intentionality criterion"; Schacter et al., 1989). Importantly, between the two versions of the task, only the instructions differed (see Appendix).

Once participants had pressed the button to report, they were then asked to provide further details of their memory (“Describe what you see or hear”). To make their entry they typed into an unconstrained text entry field. For definitions of involuntary and voluntary memory and instructions for giving the written descriptions see Appendix. Without any memory reports, the task had a duration of 9 minutes. With the memory reports the task duration increased to approximately 20 minutes. Each participant completed both voluntary and involuntary versions of the VRT, but testing order varied between participants.

Film ratings

Participants were presented with stills from each of the clips in the film and asked to “judge whether you feel the content of the video created an instant sense of impact on you personally” on a 0-9 sliding scale ranging from ‘no impact’ to ‘strong impact’ (Croucher et al., 2011), followed by rating the clips on valence and arousal, using the Self-Assessment Manikin scale (Bradley & Lang, 1994). Finally, demand characteristics were evaluated by asking participants whether they believe that playing Tetris will decrease intrusive images, using a sliding scale from -10 (extreme decrease) to +10 (extreme increase).

Diary

Participants were sent a questionnaire at the end of each day, on the day of the session and for the three days following the lab-based session. Both verbal and written instructions were given on how to complete the diary. An intrusive memory was defined as “a mental image, [that] may pop into your mind, out of the blue, and not at times when you are deliberately thinking about the film”, in line with the VRT

involuntary instructions. Participants were told to not include memories of the film that were retrieved deliberately. In the online questionnaire, participants received the following prompts: '(i) How many involuntary memories (images/sounds) about the film have popped into your mind today when you were not deliberately thinking about it; and (ii) Please describe what you pictured or heard in these memories to the best of your ability. Even if you have several involuntary memories that are the same moment of the film, please record each individual one. If you don't have any involuntary memories pertaining to the film, please record this as '0'.' Participants typed their intrusive memory descriptions into the diary in an unconstrained text field. Participants' descriptions of the intrusive memories were used to verify whether the intrusion related to the film or not.

Data analysis

We counted the total number of button-presses in the first VRT, excluding the 3.9% of the reports which could not be linked to any of the film clips (DV1). DV1 was used to test hypotheses A and B. We also counted the number of unique film details reported in the first VRT, excluding repetitions of recurring details (DV2). As is typical with recall of films, "details" were coded according to the Autobiographical Interview (Levine et al., 2002). DV2 was used for Hypotheses A-C. These data naturally include ambiguities, for example, where it is not fully clear that a response refers to any film detail, where it may refer to more than one film detail, or where it seems the participant has perceived a film detail incorrectly (e.g., a pilot participant perceived blood drops as insects). Using the Autobiographical Interview coding scheme (Levine et al., 2002), data processing classified reports as "internal details" (reports that refer to film detail), "external detail" (reports that do not refer to film detail), "errors" (reports that refer to content that was not studied in the experiment) and "other" responses. At least 20%

of the datasets, and all external detail, error and other responses, were coded independently by a second rater. Where they disagreed with the first rater, these were discussed by the entire team to reach resolution.

DV1 and DV2 were scored using the qualitative analysis software *Taguette* (<https://app.taguette.org>). Each press of the space key was tagged with the film clip number it referred to or tagged as an 'invalid' press if it did not include any reported memory of the film (3.9% of presses). DV1 was calculated by subtracting total invalid presses from total space key presses. Text within each entry was then tagged for Internal details from the film, repetitions of these Internal details, External details (non-film-related content) or errors, based on the Autobiographical Interview scoring system used for the free recall data analysis in Lau-Zhu et al. (2019). DV2 was calculated using the total number of Internal details from the film, excluding repetitions. Counts of repeated internal details or external details are provided in Supplementary Table 2. The same scoring process was conducted by a second rater for 20% of all data files, to assess inter-reliability of Internal detail scoring (ICC = 0.993 for first VRT scoring and ICC = 0.962 for second VRT scoring). Both raters were blinded to the intervention and retrieval conditions.

The seven mood rating variables (sad, happy, anxious, depressed, calm, hopeless, angry), each measured on 0 to 100 Likert scales, were averaged into a composite 'Negative Affect' score, with positively valenced items (happy and calm) being reverse scored (e.g., a rating of 10 points for 'happy' is entered as -10).

In the online diary, participants reported the total number of involuntary memories they experienced each day and entered text to describe content of those memories. To ensure that valid, accurate, numeric responses were recorded, the research team verified the memory counts for each day. If none of the written memory

content was from the film, then the count number was converted to zero (7 occurrences). Similarly, if the written content only indicated general film-related thoughts rather than involuntary memories (e.g., “Did not get flashbacks of specific scenes but got an overall sense of sadness all three times”), then the count was converted to zero (1 occurrence). In cases where non-film-related or non-intrusion descriptions were accompanied by other valid entries, one was subtracted from the inputted count (1 occurrence). With these verified counts for each day, a combined total across all days was calculated, with any missing values for a given day (Day 0: $n=22$, Day 1: $n=18$, Day 2: $n=26$, Day 3: $n=16$) treated as zeros in this addition (rather than using imputation as in previous studies).

Data exclusions

During the data collection phase, and prior to inspecting their data, we replaced participants who were clearly ineligible (e.g., arrived intoxicated), or where procedural anomalies occurred such that they did not complete the protocol appropriately (e.g. equipment malfunction, or participant did not watch the film). We pre-registered that once the stopping point was reached, we would exclude participants whose performance data are outliers and excluded participants whose performance data on DV1 and DV2 was greater than $Q3 + 1.5 \times$ the interquartile range (IQR) or lower than $Q1 - 1.5 \times$ IQR. Using these thresholds, 25 outliers were detected and replaced through additional data collection. A further six participants were excluded due to procedural issues during the initial lab session and also replaced. In these cases of additional data collection, it was ensured that the experimenter remained blinded to participants’ allocation to conditions by maintaining the approach employed during

initial data collection of identifying participants only by their anonymised ID number and inputting this number into the relevant experimental programs.

Statistical analyses

Our first two hypotheses are derived from the DRA, but evidence for the null could support a unitary account of trauma memory, while the opposite is true for Hypothesis D. Because the strength of the evidence for both the experimental hypothesis and the null hypothesis were of interest, we used Bayesian statistics to examine our results. All statistical analyses used the package *BayesFactor* (Morey & Rouder, 2018).

Based on pilot data, we expected DV1 to range between 5-32 key presses and DV2 to range between 8-86, and for both DV1 and DV2 to have an approximate Gaussian distribution, such that the count data can be treated as continuous data and parametric statistics applied. We therefore tested hypotheses A and B with a 2-way ANOVA using the *anovaBF* function. These hypotheses correspond to different 2X2 interactions within this design (see Table 1) and were evaluated against models with main effects only. In the presence of interactions, follow-up tests of simple effects were performed on subsets of the data. Hypothesis C corresponds to the simple effect of emotion in the voluntary/control intervention condition and was examined using a 1-way ANOVA. Hypothesis D was tested with *correlationBF* from the same R package.

That said, it was also possible that count numbers for DV1 and DV2 could be low, skewed, and zero-inflated. In this case, we pre-registered to switch to implementing the same ANOVA contrasts, but within the *brm* function of the *brms* package (Bürkner, 2017), which enables various linking functions to handle such count data. Specifically, we planned to replicate the *brm* models used to handle similar count

data by Ramineni et al. (2023; see also <https://osf.io/m5cvj/>), using model comparison to determine the best model, from which we estimated the corresponding Bayes Factors as above. The code we pre-registered to use to apply and compare *brm* models can be found within the folder 'BRMS code' under the following link: https://osf.io/cms7n/files/s2g9h?view_only=0fa9878af33b40de9a54581c71af5217. When fitting *brms* models in the reported analyses, diffuse normal priors $N(0,10)$ were initially employed for all regression coefficients, based on the approach employed in Ramineni et al. (2023). As sensitivity analyses, more constrained normal priors $N(0,1)$ were also implemented for all regression coefficients.

The following analyses were conducted for the purpose of manipulation checks: (i) a numerical comparison between the number of diary intrusions and involuntary memory reports from the version of the VRT used here and the earlier version of the VRT in previous studies to indicate the extent to which our sample was similar in their response in the trauma condition (ii) a comparison between the baseline mood ratings in the trauma vs. the neutral groups, and (iii) a comparison between pre- and post-film mood rating in the trauma vs. the neutral groups, to ensure that the neutral film did not induce negative mood in participants.

We planned to conduct a number of secondary analyses to examine additional questions, all using functions from the *BayesFactor* package. Although our diary measure was taken only over the course of 4 days (once daily), and uses an electronic method, we believed that it would still be of interest to use the number of intrusions reported in the diary as DV to examine Hypothesis B, and to validate the VRT by examining whether VRT and diary measures of involuntary memories are correlated across participants.

Results

Stopping data collection

Following the pre-registered stopping rule, a stop/go script was run examining Bayesian evidence for Hypotheses A and B and for the null effect using *anovaBF* of DV2. At N=240, the script indicated data collection should stop. This was due to evidence for the null for both interaction effects ($BF_{01} > 3$). As stated above, the pre-registered plan was to use the thresholds of $BF_{01} > 6$ (or $BF_{10} > 6$) for these tests, however due to a technical error $BF_{01} > 3$ was instead used to evaluate the evidence for the null effects.

Manipulation checks

Mood ratings

Using the mood rating questionnaires administered at Baseline (T1), Post-Film (T2) and Post-Intervention (T3) negative affect scores were calculated at each timepoint, as shown in Supplementary Figure 1. At T2, participants in the Negative condition showed increased negative affect ($M = 34.66$, $SD = 16.87$), whereas participants in the Neutral condition showed a slight decrease ($M = 13.19$, $SD = 8.52$), with extreme Bayesian evidence for this difference, $BF_{10} = 2.53 \times 10^{24}$. In a secondary analysis, a Bayesian independent samples *t*-test indicated that there was inconclusive evidence for or against a difference in negative affect at T1 between the Negative ($M = 19.56$, $SD = 13.37$) and Neutral groups ($M = 16.64$, $SD = 10.62$), $BF_{01} = 1.35$.

Post-film questionnaire

Self-reported levels of attention to the film, percentage of time looking away from the film, and degree of immersion in the film resembled that in previous studies and were similar in the Negative and Neutral groups (Attention: Negative $M = 92.91$, $SD = 6.90$, Neutral $M = 92.68$, $SD = 7.82$; Time Looking Away: Negative $M = 5.88$, $SD = 12.34$, Neutral $M = 4.16$, $SD = 12.58$; Immersion: Negative $M = 81.83$, $SD = 14.05$, Neutral $M = 80.53$, $SD = 14.13$). The Negative group reported greater film-induced distress ($M = 68.27$, $SD = 23.11$) than the neutral group ($M = 3.17$, $SD = 12.04$).

Film ratings

Self-reported ratings of the impact, arousal and valence for each of the film clips were averaged and compared for each of the two Emotion conditions (see Supplementary Figure 2), using Bayesian t -tests. There was extreme evidence for higher ratings of impact in the Negative condition ($M = 5.60$, $SD = 1.35$) relative to the Neutral condition ($M = 4.08$, $SD = 1.57$), $BF_{10} = 1.43 \times 10^{11}$. There was also extreme evidence for lower ratings of valence in the Negative condition ($M = 3.33$, $SD = 0.84$) compared to the Neutral condition ($M = 5.81$, $SD = 0.69$), $BF_{10} = 2.23 \times 10^{64}$, and for higher ratings of arousal in the Negative ($M = 5.21$, $SD = 1.27$) compared to the Neutral condition ($M = 3.87$, $SD = 1.11$), $BF_{10} = 9.32 \times 10^{12}$.

Demand ratings

Expectations regarding the effect of the visuospatial intervention on involuntary memories were measured using a scale from -10 (reduction) to +10 (increase). Overall, pooling across Intentionality and Emotion conditions, the Visuospatial group ($M = -1.05$, $SD = 4.89$) expected more of a decrease than the

Control group ($M = -0.28$, $SD = 3.82$), however an exploratory Bayesian t-test found evidence against a difference in demand ratings ($BF_{01} = 2.96$).

Diary reports

The percentage match of diary entries to the film was 99.20% (7/878 completed reports did not match film content). The combined total number of involuntary memories reported across the four days of the diary was calculated in each of the Emotion and Intervention condition. On average, the total number of diary involuntary memories was higher in the Negative condition (Control $M = 4.23$, $SD = 3.98$; Visuospatial $M = 4.53$, $SD = 5.59$) than in the Neutral condition (Control $M = 3.80$, $SD = 5.44$; Visuospatial $M = 2.48$, $SD = 3.16$).

In a secondary analysis of the effects of Emotion and Intervention on total diary intrusions, there was inconclusive evidence for a main effect of Emotion ($BF_{10} = 1.02$), moderate evidence against a main effect of Intervention ($BF_{01} = 5.18$) and anecdotal evidence against an interaction effect ($BF_{01} = 2.30$).

In a secondary analysis, the relationship between total diary intrusions and DV1 in the Involuntary VRT was also examined, pooling across Intervention groups. There was substantial evidence for a positive correlation in the Negative condition ($r = 0.32$, $BF_{10} = 5.13$) and anecdotal evidence against any correlation in the Neutral condition ($r = 0.08$, $BF_{01} = 2.88$).

Memory reports in the Vigilance Retrieval Task

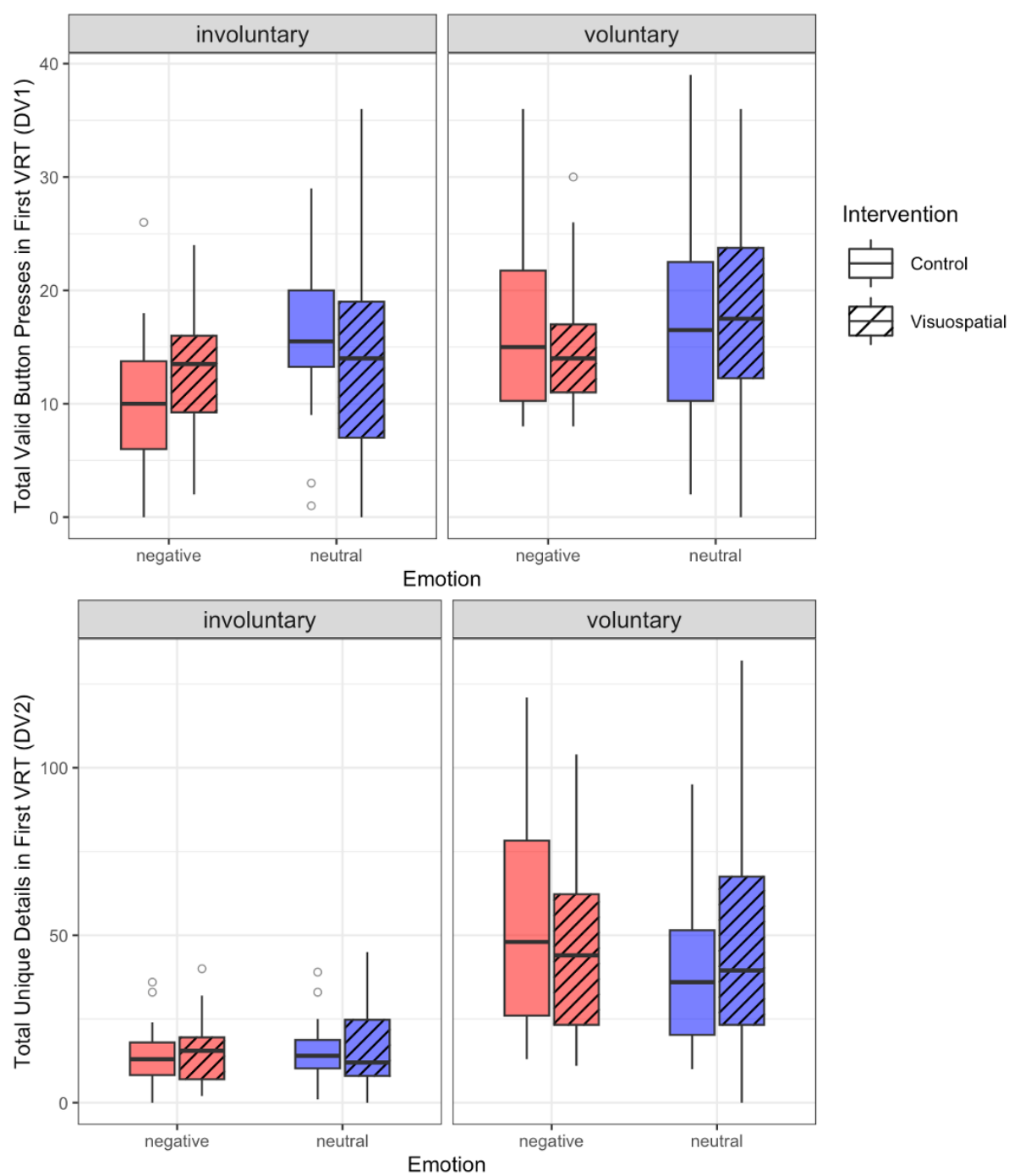
As the primary dependent variables derived from the VRT are both count data, Figure 3 visualises comparisons of each experimental group based on the medians and interquartile ranges of these variables. The figure suggests that the

visuospatial intervention did not have the anticipated effect on involuntary memory in the Negative group.

1028

Figure 3

Lab-based Involuntary and Voluntary Memory Counts on the Vigilance Retrieval Task.



1031

Note. Data are derived from the first Vigilance Retrieval Task (VRT) completed by each participant (i.e., either voluntary or involuntary). Boxes indicate the group-level median and interquartile range with N=30 per group. Since there were outliers in the final sample even after replacing outliers, circles indicate those who, exceeded the third quartile by 1.5 times the interquartile range.

Given the potential skew of these count variables (see Supplementary Figure 3), *brms* model comparison was used to determine the optimal model distribution to apply. Due to the negligible prevalence of zeros across both outcome variables (3/240), zero-inflated models were not fitted. Thus, for tests of Hypotheses A–C, candidate models were fitted using Gaussian, Poisson, and Negative Binomial families.

For tests using DV1, Bayes factors in model comparisons consistently favoured the Gaussian models, indicating the values approximated a normal distribution. However, for tests using DV2, the Bayes factors consistently favoured the Negative Binomial model. Thus, for tests using DV1, *anovaBF* results were retained, while for DV2, Negative Binomial *brms* models are also reported.

Tests of Hypothesis A

Examining Figure 3, it is clear that the data does not align with Hypothesis A for an interaction between intentionality and intervention, because unexpectedly, the visuospatial intervention did not decrease intrusions compared to the control intervention. In the Involuntary conditions DV1 and DV2 for the Negative film are both numerically higher, rather than lower, in the Visuospatial intervention group compared to the Control group, while for the equivalent after the Neutral film they are numerically lower.

A Bayesian ANOVA on DV1 in the Negative condition showed extreme evidence for a main effect of Intentionality ($BF_{10} = 362.54$), with more memories reported in the Voluntary than Involuntary condition, as expected, but moderate evidence against a main effect of Intervention ($BF_{01} = 4.93$). Crucially, there was no evidence for or against an Intentionality $\times$ Intervention interaction ($BF_{10} = 1.18$).

For DV2, the Bayesian Negative Binomial regression models (see above) again showed extremely strong evidence for a main effect of Intentionality ($BF_{10} = 2.39 \times 10^{15}$ using $N(0,10)$ priors; $BF_{10} = 1.16 \times 10^{16}$ using $N(0,1)$ priors), but moderate to strong evidence against a main effect of Intervention ($BF_{01} = 90.91$ using $N(0,10)$ priors; $BF_{01} = 8.93$ using $N(0,1)$ priors). Crucially, there was evidence against the interaction ($BF_{01} = 24.39$ using $N(0,10)$ priors; $BF_{01} = 3.01$ using $N(0,1)$ priors).

Tests of Hypothesis B

Since the visuospatial intervention did not decrease intrusions compared to the control intervention, the data also did not align with hypothesis B that it would do so more in the trauma compared to the neutral condition. A Bayesian ANOVA to test Hypothesis B in the Involuntary condition using DV1 showed strong evidence for a main effect of Emotion ($BF_{10} = 15.81$), with more intrusions in the Neutral than Negative condition, contrary to expectations. There was moderate evidence against a main effect of Intervention ($BF_{01} = 5.13$), and no evidence for or against an Emotion $\times$ Intervention interaction ($BF_{10} = 1.07$).

For DV2, there was moderate to strong evidence against a main effect of Emotion ($BF_{01} = 62.50$ using $N(0,10)$ priors; $BF_{01} = 6.25$ using $N(0,1)$ priors), unlike DV1 above. There was also moderate to strong evidence against a main effect of Intervention ($BF_{01} = 55.56$ using $N(0,10)$ priors; $BF_{01} = 5.49$ using $N(0,1)$ priors), and

against any interaction ($BF_{01} = 41.67$ using $N(0,10)$ priors; $BF_{01} = 4.44$ using $N(0,1)$ priors).

Tests of Hypothesis C

Figure 3 shows that for the Control group in the Voluntary condition, there was no suggestion that DV2 was lower in the Negative than Neutral condition, as would be expected by the DRA. Indeed, Bayesian Negative Binomial regression models showed evidence against an effect of Emotion ($BF_{01} = 10.53$ using $N(0,10)$ priors; $BF_{01} = 1.12$ using $N(0,1)$ priors), providing evidence against Hypothesis C.

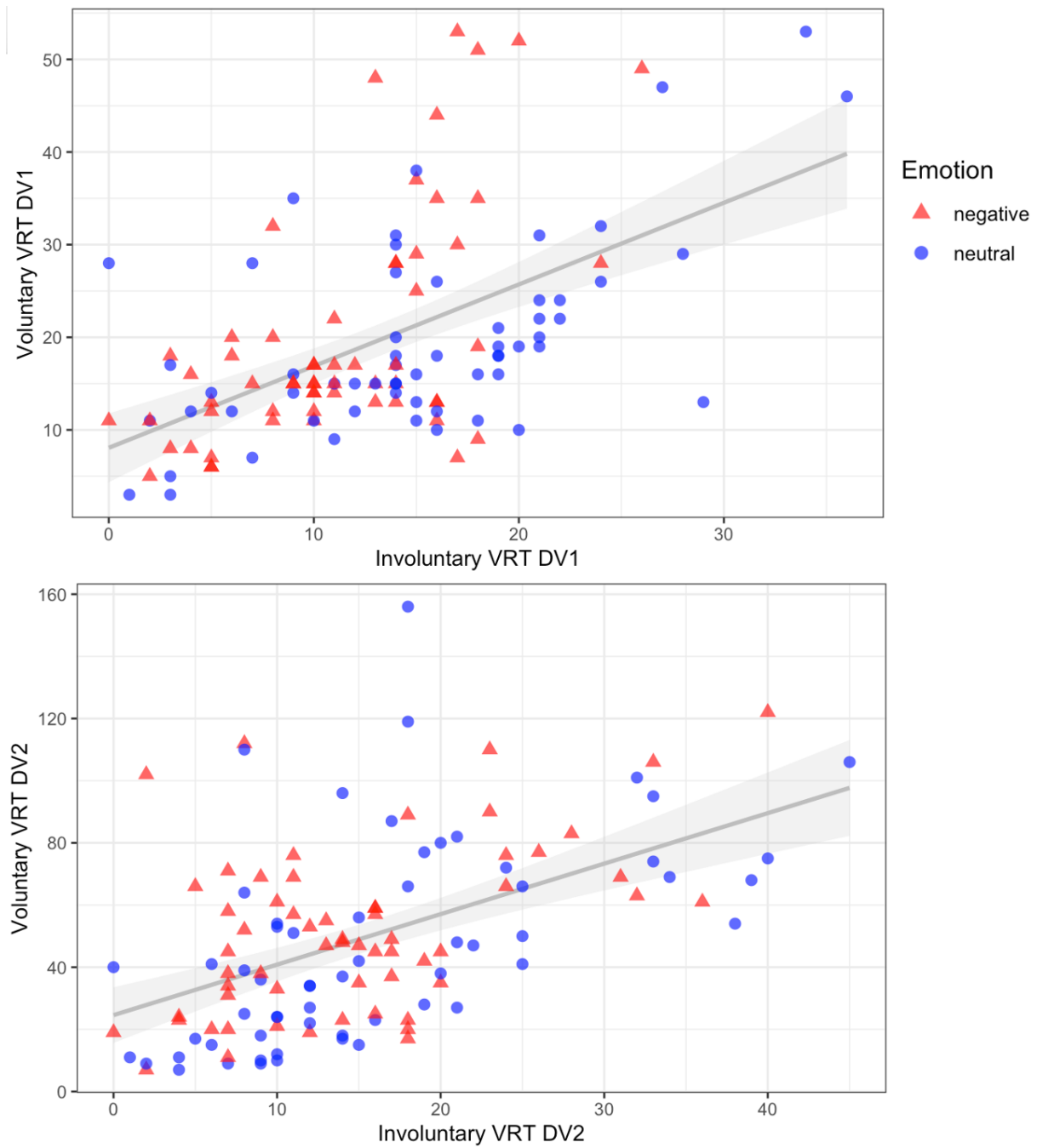
Given the unexpected null effect of the intervention in this paradigm, an exploratory analysis increased statistical power by pooling data from both intervention groups to assess the effect of Emotion on Voluntary recall. Within this pooled dataset, DV2 was numerically higher in the Negative condition ($M = 49.18$, $SD = 28.96$) than the Neutral condition ($M = 44.72$, $SD = 31.56$), however there was moderate to strong evidence against an effect of Emotion ($BF_{01} = 58.82$ using $N(0,10)$ priors; $BF_{01} = 6.10$ using $N(0,1)$ priors).

Tests of Hypothesis D

To test Hypothesis D, Bayesian correlations were examined between memory outcomes in the Involuntary VRT and subsequent Voluntary VRT in the sample who completed the involuntary test first. Data were combined across Intervention and Emotion conditions. Both when examining DV1 ($r = 0.54$, $BF_{10} = 7.06 \times 10^7$) and DV2 ($r = 0.51$, $BF_{10} = 6.07 \times 10^6$), the two memory tests were highly positively correlated (Figure 4).

Figure 4

Correlations between involuntary memory and subsequent voluntary memory



1110

1111 *Note.* Data only from the Involuntary-first condition for each Emotion group (N=60 per

1112 group), collapsing across the Intervention condition. Points indicate individual participants.

1113 Regression lines are displayed with standard error shading.

1114

1115 **Exploratory analyses**

1116 To examine the potential effects of VRT test order, a series of exploratory
1117 2x2x2 between-subjects Bayesian ANOVAs tested main and interaction effects of
1118 Emotion, Intentionality (i.e. test order), and Intervention on two dependent variables
1119 — DV1 in the Involuntary test and DV2 in the Voluntary test, the measures of greater
1120 interest to their respective fields.

1121 For DV1, the main effects of Intentionality ($BF_{01} = 3.33$) and Intervention (BF_{01}
1122 $= 6.41$) moderately supported the null, though there was strong evidence for a main
1123 effect of Emotion ($BF_{10} = 25.77$). All two-way interactions – Emotion x Intentionality
1124 ($BF_{01} = 3.75$), Emotion x Intervention ($BF_{01} = 2.31$), and Intentionality x Intervention
1125 ($BF_{01} = 5.26$) – and the three-way interaction ($BF_{01} = 2.14$) also provided anecdotal to
1126 moderate support for the null. For DV2, there was anecdotal to moderate evidence
1127 for the null for all main effects – Emotion ($BF_{01} = 3.89$), Intentionality ($BF_{01} = 5.75$),
1128 and Intervention ($BF_{01} = 6.13$) — as well all two-way interactions – Emotion x
1129 Intentionality ($BF_{01} = 5.29$), Emotion x Intervention ($BF_{01} = 2.56$), and Intentionality x
1130 Intervention ($BF_{01} = 5.24$) – and the three-way interaction ($BF_{01} = 2.01$).

1131 Critically, neither the main effect of Intentionality nor any of its interactions
1132 reached evidential significance for either dependent variable, indicating an absence
1133 of test order effects. These results suggest that order effects do not contaminate
1134 current practice, such that it may be viable to employ a within-subject design, with
1135 DV1 as a measure of involuntary memory and DV2 as a measure of voluntary
1136 memory.

1137 As a further exploratory analysis, we conducted the analyses we pre-
1138 registered for Experiment 2 on data from Experiment 1. These analyses focused on
1139 the effect of Intervention on Involuntary memories in the Negative condition only,

using one-sided Bayesian t -tests (Visuospatial < Control). All tests provided support for the null hypothesis (all $BF_{01} > 6$), including for DV1 ($BF_{01} = 9.52$), DV2 ($BF_{01} = 6.34$), and total diary intrusions ($BF_{01} = 6.53$).

Discussion

The diary reports in participants who took part in the negative condition indicate that the trauma film successfully induced intrusive memories. In the negative condition, the number of intrusions reported during the involuntary VRT was positively correlated with the number of intrusions reported in the diary, supporting the VRT as a laboratory operationalisation of conditions which give rise to involuntary memory in the real world. The self-reported mood data also shows that the manipulation of emotion in this experiment was successful. This is evident in that compared to the neutral group, the negative group reported more negative affect after the film, rated the film as more distressing, and rated the individual clips from the film as more impactful, negative, and arousing. Despite these successful manipulation checks, none of our predictions that were derived from the DRA were supported

Our experimental design had assumed that visuospatial intervention would interfere with involuntary memory in the negative film condition. This assumption was based on the balance of evidence that was reviewed above and supported by recent meta-analytic evidence (Varma et al., 2024, but see exceptions mentioned previously with null and mixed findings). Indeed, the memory dissociation findings of Lau-Zhu et al. (2019, 2021) was the impetus for the research presented here and the reason we hypothesized that this visuospatial intervention would decrease involuntary memory more than voluntary memory (Hypothesis A) and negative involuntary memory more than neutral involuntary memory (Hypothesis B). However, the comparison between the visuospatial interference intervention and the control intervention in the negative

film condition indicated that there was moderate evidence for a null interference effect. This finding decreases confidence in the claim that voluntary and involuntary memory can be functionally dissociated. We discuss the implications of these results in the General Discussion.

Our data collection process was blinded, so it was only when data collection was terminated and we viewed the data for the first time that we realised that the visuospatial intervention failed to decrease involuntary memories in the negative condition. In hindsight, we should have included a test for the presence of this effect in our stopping rule, as a condition of continued data collection. Because in this dataset we did not observe a selective interference effect, Hypotheses A and B, which defined the stopping rule, were not supported. This was despite the fact that we conducted some pilot testing so trainee researchers could get feedback and monitor how participants responded, although experimenter training here was less detailed as in Lau-Zhu et al. (2019). We reviewed experimenter training (see McConnell et al., 2026) and enhanced it in Experiment 2.

The unexpected null effect of the visuospatial intervention does not impinge on Hypothesis C, which like the others, was also derived from the DRA. Hypothesis C was conditionalized on finding diary reports of involuntary memory in the negative condition, because according to the DRA, the presence of intrusions suggests that some quasi-pathological encoding has occurred, at least at the level of the group. Having satisfied this condition, we next tested the prediction derived from this theory that voluntary memory for negative films will be impaired. The results, tested only in participants who conducted the voluntary memory test first, contradicted this hypothesis, with substantial evidence that voluntary memory for negative films did not differ in the number of details from that of neutral films.

While the finding that participants in the negative condition, who reported a rise in negative affect and emotional distress, did not recall more than participants in the neutral condition (in either voluntary or involuntary measures) may initially appear inconsistent with prior findings of emotionally enhanced memory, it aligns with predictions of unitary accounts of memory. In our previous publications, we have frequently observed an emotional list composition effect: that emotion increases memory when emotional and neutral items are intermixed at study and test, but not when such items are blocked, i.e., when emotional stimuli are studied and recalled separately from neutral stimuli (Talmi, Luk et al., 2007; Talmi & McGarry, 2012; Barnacle et al., 2017). This effect was observed by other groups (Hadley & MacKay, 2006; Garcia, Ritchey & Kensinger, 2023), although failed replications have also been reported, where emotion enhanced memory regardless of list composition (Zarubin et al., 2020; Baraly et al., 2019). eCMR and CMR3 are unitary models of memory that account for the emotional list composition effect (Talmi et al., 2019; Cohen & Kahana, 2022). They suggest that although emotional experiences always draw processing resources and therefore have a ‘stronger’ memory trace (which in the models is operationalised as an increased learning rate), this only gives them a mnemonic advantage when the memory test that takes place at test includes weaker items, those being neutral experiences in mixed list conditions.

Given the balance of evidence that underlies the effects of visuospatial interventions to reduce intrusive memories both in the laboratory (Asselbergs et al., 2024; Badawi et al., 2020; Deeproose et al., 2012; Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020; Lau-Zhu et al., 2019, 2021) and in the clinic (e.g., Iyadurai et al., 2018; Kanstrup et al., 2024), the null effect of the visuospatial intervention here is more similar to other studies that have either found no reduction

in intrusive memories (e.g., Asselbergs et al., 2018; Brühl et al., 2019; Brennen et al., 2021; Broughill et al., 2025; Matura et al., 2025), or only on immediate but not longer-term measures (Wessel et al., 2025). It is natural to wonder what aspect of the protocol of Experiment 1 resulted in the failure to replicate some of the previous studies. Experiment 2 is the result of extensive discussions and represents our best attempt to replicate the previously observed effects of visuospatial interference on lab-based intrusions using a refined methodological approach closer to Lau Zhu et al. (2019).

Experiment 2

Experiment 2 was designed to give us the best chance of replicating previous findings by Lau-Zhu et al. (2019, Experiment 2) that a visuospatial intervention decreases involuntary memories within the trauma film paradigm. For this purpose, we focused on two of the eight conditions employed in Experiment 1, comparing only two groups, both of which encoded only the negative film clips from Experiment 1. One group received a visuospatial intervention, and the other served as a control group. We retained the same dependent measures from Experiment 1, i.e. the involuntary VRT and diary, but did not include the voluntary VRT.

The protocol of Experiment 2 further differed from the protocol of Experiment 1 in the following ways. First, the verbal control task employed in Experiment 1 (listening to a political podcast) was replaced with a passive rest period, to more closely align with Lau-Zhu et al. (2019). This change addressed the possibility that our previous control task was not sufficiently neutral in topic and may have unintentionally elicited imaginal cognition. Second, while in Experiment 1 the Visuospatial task instructions and practice took place after the reminder cues, all participants in Experiment 2 completed a practice session of Tetris with instructions emphasising mental rotation, before film viewing. This reduces the interval between delivering reminder cues and

beginning the intervention and more directly aligned with the protocol used in Lau-Zhu et al. (2019). Third, the structured filler period between the film and the reminder cues was increased from 5 min in Experiment 1 to 30 min in Experiment 2, replicating the 30 min filler period in Lau-Zhu et al. (2019).

Additionally, in Experiment 2, the experimenter was fully blinded to participants' allocation to the intervention or control condition, with post-reminder task instructions administered by a computer program without the experimenter present. This is in contrast with both the present Experiment 1 and Lau-Zhu et al. (2019), in which the experimenter directly administered the intervention, creating the potential for experimenter-related bias in subsequent memory measures. Supplementary Table 1 lists the remaining procedural similarities and differences between the current Experiment 2 and Experiment 2 of Lau-Zhu et al. (2019).

Finally, we reflected on the training experimenters typically receive in experimental psychopathology research and considered good practice (McConnell et al., 2026). Unlike Experiment 1, which involved only a brief period of experimenter training, more extensive experimenter training was employed prior to Experiment 2, across several expert-guided sessions online and one in person, covering the delivery of trauma film viewing instructions, mental rotation instructions and involuntary memory measurement instructions.

Methods

Aiming to replicate the findings of Lau-Zhu et al. (2019) Experiment 2, we hypothesised that involuntary memories would be lower in the Visuospatial (Reminders+Tetris) than Control (Reminders+Rest) condition, as measured by DV1

(number of involuntary VRT memories), DV2 (number of involuntary VRT memory details) and Diary Total (combined total diary intrusions).

Transparency and openness

This experiment was pre-registered on OSF (<https://osf.io/cg2t5>) with data and analysis code available there. In Supplementary Table 1, we delineate all known differences in methodology between the present experiments and the experiment we aimed to replicate.

Participants

We employed a Bayesian sequential design (Schönbrodt & Wagenmakers, 2018), beginning with an initial sample of $n = 20$ per group and adding data in batches of 10 per group. Data collection continued until reaching strong evidence for the alternative hypothesis ($BF_{10} > 6$) or the null hypothesis ($BF_{01} > 6$), or at a maximum of $n = 60$ per group. The stopping decision was based only on the number of memories reported in the Involuntary VRT (DV1).

A large effect size was expected, based on previous research cited above (e.g., Lau-Zhu et al., 2019, Experiment 2 VRT: $d = 0.96$). To evaluate our sequential design, Monte Carlo simulations were conducted in R. In each of 5000 iterations, data for two independent groups were sampled in batches of 10, starting from a minimum of $n = 20$ per group and ending either when the Bayes Factor reached the predefined decision thresholds or when a maximum of $n = 60$ was reached. Bayes factors were calculated using the *ttestBF* function from the *BayesFactor* package (Morey & Rouder, 2018), with default priors and a one-tailed null interval, reflecting the directional hypothesis that DV1 will be lower in the Visuospatial group than the Control group.

Under a simulated large effect size ($d = 0.80$), the median final sample size was the minimum of 20 per group, with 97.72% of simulations reaching $BF_{10} > 6$, while only 2.28% reached the maximum sample size without conclusive evidence at $BF_{10} > 6$ or $BF_{01} > 6$. The rate of incorrect conclusions in favour of the null ($BF_{01} > 6$) was 0.00%, indicating negligible risk of false negatives in the case of a true effect. To assess the potential to detect a true null effect, simulations were also run with the effect size set to zero ($d = 0.00$). In these null simulations, the median final sample size was the maximum 60 per group, with 51.00% of simulations producing strong evidence for the null ($BF_{01} > 6$), while 46.58% were inconclusive at $BF_{10} > 6$ or $BF_{01} > 6$ and only 2.42% incorrectly supported the alternative ($BF_{10} > 6$). This demonstrates that our design had potential to support the null hypothesis if true, with minimal scope for false positives.

To minimise any potential bias during data analysis, at each stopping point, the personnel who conducted the initial data analysis utilised an automated analysis script, which provided them only with a 'stop' or 'go' answer, but not the BF associated with the hypothesis (Sadibolova & Terhune, 2025).

As in Experiment 1, participants were recruited from the local community through the Cambridge SONA platform. The inclusion and exclusion criteria were the same as Experiment 1.

The automated analysis script did not indicate stopping at any of the preregistered stopping points. Therefore, testing continued to the maximum sample size of $N=120$. Data exclusions followed the pre-registered plan. During the course of testing, three participants were replaced due to technical issues during data collection. Upon reaching the maximum preregistered sample size outlier checks were conducted and 8 participants were replaced whose performance data were

outliers with respect to their group. The final sample included 60 participants in the Visuospatial group (44 females and 16 males; age $M = 29.98$, $SD = 13.94$) and 60 in the Control group (40 females and 20 males; age $M = 28.75$, $SD = 12.03$).

Data analysis

The following analyses were preregistered. To assess our key hypothesis, one-tailed Bayesian t -tests were conducted comparing the Visuospatial and Control groups across DV1 (number of involuntary memories), DV2 (number of involuntary memory details) and Diary Total (combined total diary intrusions). All DVs were computed in the same way as Experiment 1 and the same statistical approach was followed to inspect the distribution and decide on the appropriate statistical model for these count data.

The following pre-registered secondary analyses were also conducted. As a manipulation check, we conducted two-sided Bayesian t -tests comparing negative mood before and after the trauma film. To gain a better understanding of the VRT, we also examined the following correlations using the *correlationBF* function: between the number of involuntary memories in the VRT and the total number of intrusions reported in the diary, separately for DV1 and DV2; and between the number of involuntary memories and the change in negative mood before and after the trauma film, separately for DV1, DV2 and total diary intrusions.

The processing of VRT data followed the same procedures as described in the Experiment 1 results section. 4.7% of presses were invalid. Inter-rater reliability of Internal detail scoring was excellent ($ICC = 0.98$).

Diary data was processed as in Experiment 1. If none of the written memory content was from the film, then the count number was converted to zero (4

occurrences). Similarly, if the written content only indicated general film-related thoughts rather than involuntary memories, then the count was converted to zero (3 occurrences). In cases where non-film-related or non-intrusion descriptions were accompanied by other valid entries, one was subtracted from the inputted count (2 occurrences). When calculating Diary Total, any missing values for a given day (Day 0: n=3, Day 1: n=3, Day 2: n=3, Day 3: n=8) were again treated as zeros in this addition.

Materials and procedure

Experiment 2 employed the same materials and procedure as in the Negative and Involuntary conditions of Experiment 1, with the overall timeline displayed in Figure 2 and any minor differences detailed above and in Supplementary Table 1.

Allocation to the Visuospatial or Control condition was completed without the experimenter present, by a computer program created using Gorilla Experiment Builder (Anwyl-Irvine et al., 2020; www.gorilla.sc). The experimenter entered the participant ID (which determined pseudo-random allocation) and then left the room. The participant then received written instructions to either play Tetris with mental rotation for 10min or to passively rest for 10min. In both cases, this was timed by the computer program, with a beeping sound playing to indicate when the 10min had ended. The program then indicated to the participant that their "brief task" has ended, and they should notify the experimenter. In the Control condition, a countdown timer was displayed on screen, and participants were instructed to rest and attend to this timer. In the Visuospatial condition, the only stimulus visible was the Tetris game itself, to minimise distraction.

Results

Manipulation checks

Mood ratings

A comparison of Baseline (T1) and Post-Film (T2) negative affect scores was conducted as a secondary analysis (see Supplementary Figure 1). There was a mean increase in negative affect of 23.42 points pooled across both intervention groups, with a two-sided paired Bayesian t -test providing extreme evidence for this difference, $BF_{10} = 1.95 \times 10^{23}$.

Post-film questionnaire

Self-reported levels of attention to the film, percentage of time looking away from the film, and degree of immersion in the film were again similar in the Visuospatial and Control groups (Attention: Visuospatial $M = 95.38$, $SD = 6.25$, Control $M = 94.85$, $SD = 6.69$; Time Looking Away: Visuospatial $M = 5.57$, $SD = 13.44$, Control $M = 3.82$, $SD = 7.21$; Immersion: Visuospatial $M = 86.53$, $SD = 14.43$, Control $M = 83.32$, $SD = 14.36$). Ratings of film-induced distress (Visuospatial $M = 75.72$, $SD = 15.86$, Control $M = 72.23$, $SD = 22.47$) were also comparable to those in the Negative film group in Experiment 1.

Memory reports in the Vigilance Retrieval Task

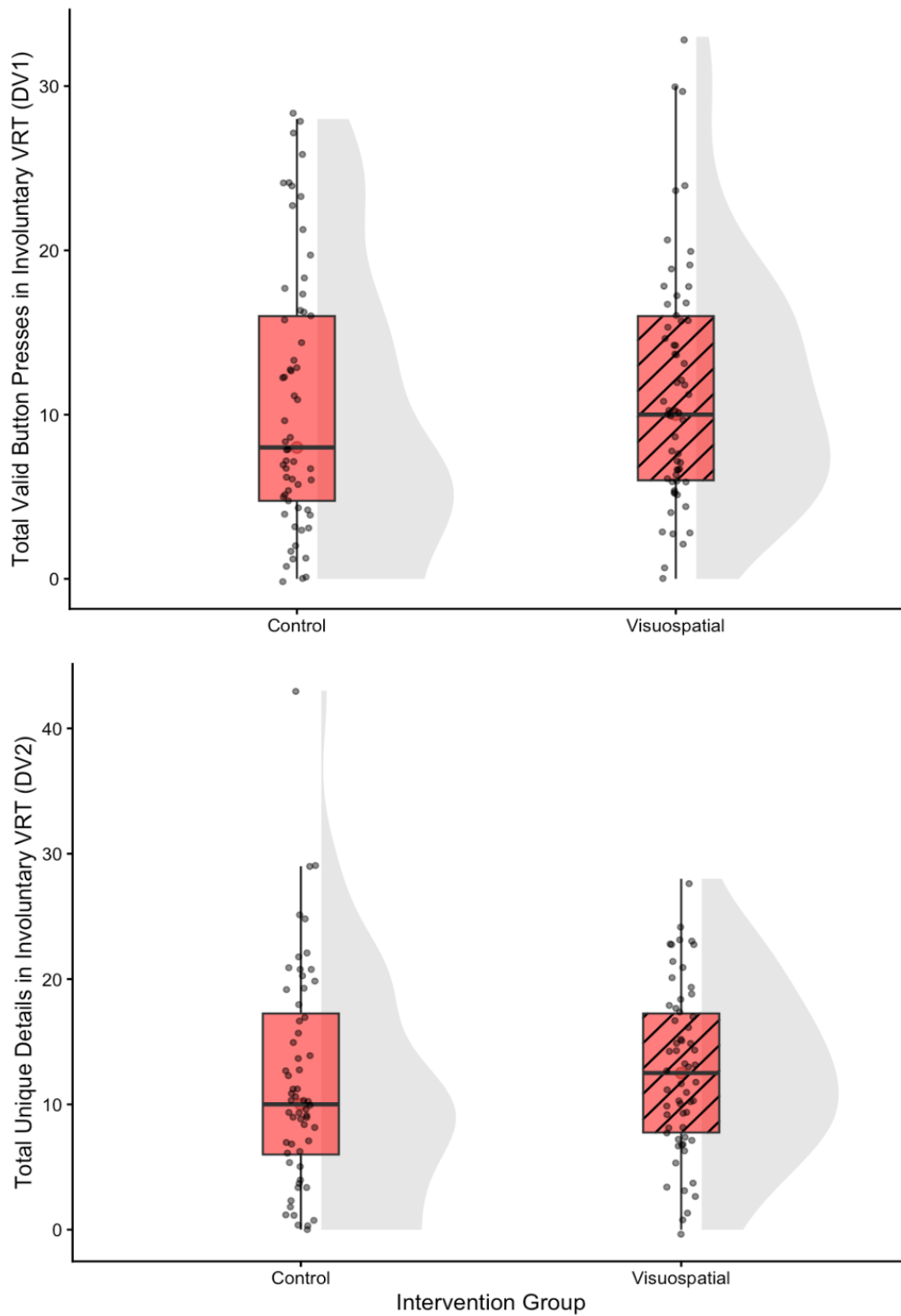
The values and distributions of DV1 and DV2 in the two intervention groups are displayed in Figure 5. For the key sequential design stopping decision test, a one-sided Bayesian t -test comparing DV1 in the Visuospatial group and Control group provided substantial support for the null hypothesis, $BF_{01} = 7.69$, exceeding the planned stopping criterion of $BF_{01} > 6$. Similarly, a one-sided Bayesian t -test

1393 comparing DV2 in the Visuospatial group and Control group also provided
1394 substantial support for the null hypothesis, $BF_{01} = 7.34$.

1395

1396 **Figure 5**

1397 *Lab-based Involuntary Memory Counts from the Vigilance Retrieval Task (VRT)*



1398

1399 *Note.* Top: Boxes indicate the group-level median and interquartile range. Probability

1400 density plots reflect group-level distributions with N = 60 per group. Points reflect

1401 individual datapoints.

1402

Since there were just 4 participants who reported zero intrusions (3.3% of participants), zero-inflated candidate models were not included in model comparisons between Gaussian, Poisson and negative binomial models. Based on these comparisons, Bayesian Negative Binomial models were selected to examine the effect of intervention group on both DV1 and DV2.

For DV1, this provided anecdotal evidence against an effect of intervention ($BF_{01} = 2.50$ using $N(0,10)$ priors; $BF_{01} = 2.44$ using $N(0,1)$ priors). For DV2, there was also anecdotal evidence against an effect of intervention ($BF_{01} = 2.17$ using $N(0,10)$ priors; $BF_{01} = 2.13$ using $N(0,1)$ priors).

Diary data analyses

The percentage match of diary entries to the film was 99.14% (4/463 completed entries did not match film content). The combined total number of involuntary memories reported across the four days of the diary (Diary Total) was highly similar across the Visuospatial group ($M = 3.97$, $SD = 3.80$) and Control group ($M = 3.88$, $SD = 3.89$). Consistent with this description, a one-sided Bayesian t -test comparing the effects of group on Diary Total provided support for the null hypothesis, $BF_{01} = 5.61$.

Diary Total distributions were positive skew characteristic of count data, and there were 24 instances of zero diary intrusions reported (20% of participants). Generalized Bayesian model comparison favoured a Negative Binomial model, from which a one-sided test provided little evidence for or against an effect of intervention ($BF_{01} = 1.18$ using $N(0,10)$ priors; $BF_{01} = 1.19$ using $N(0,1)$ priors).

Secondary correlational analyses

A Bayesian correlation was conducted to examine the relationship between our key dependent variables of DV1 and Diary Total, finding extreme evidence for a positive association ($r = 0.42$, $BF_{10} = 9638.81$). A similar relationship was observed between DV2 and Diary Total, with strong evidence for a positive correlation ($r = 0.27$, $BF_{10} = 14.49$).

Bayesian correlations were also conducted to examine potential relationships between increases in Negative Affect from T1 to T2 and involuntary memory measures. These analyses supported the null hypothesis of no association between change in negative affect and involuntary memories, across DV1 ($r = -0.05$, $BF_{01} = 4.02$), DV2 ($r = -0.08$, $BF_{01} = 3.34$) and Diary Total ($r = 0.11$, $BF_{01} = 2.34$).

Discussion

While the principal aim of Experiment 2 was to replicate previous findings that a visuospatial intervention decreases involuntary memories of analogue trauma (Asselbergs et al., 2024; Deeproose et al., 2012; Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020; Lau-Zhu et al., 2019, 2021; see also Varma et al., 2024 for a meta-analysis featuring 11 experiments using this intervention, and other null findings mentioned above), the results instead reaffirmed those of Experiment 1. The negative film once again elicited a clear rise in negative affect, as well as comparable levels of lab- and diary-based involuntary memories to previous trauma film research. Validating the vigilance-retrieval task we employed, intrusions within this lab-based task were again predictive of subsequent diary intrusions. Yet, despite extensive efforts to match the methodology of this experiment to that of previous work (Lau-Zhu., 2019, see Supplementary Table 1), including by using a passive

1453 rather than active control condition, we still observed no evidence that visuospatial
1454 intervention decreased involuntary memories. Tests of all three dependent measures
1455 (DV1, DV2 and Diary Total) supported the null hypothesis.

1456 The Bayesian sequential design, which reached the maximum planned
1457 sample size of N=120, ensures this non-replication is extremely unlikely to be
1458 attributable to Type II error. Confidence in the present findings is further bolstered by
1459 the use of experimenter-blinded allocation to the intervention and rest conditions, a
1460 form of experimental control absent from the experiments of Lau-Zhu et al. (2019).

1461 The impetus for Experiment 2 had been to produce an interference effect on
1462 lab-based involuntary memory of a negative film, in order to subsequently test our
1463 initial DRA-derived hypotheses regarding the *selectivity* of such interference (i.e.,
1464 whether the interference would extend to voluntary memory and to neutral memory).
1465 However, our results suggest this is not a feasible approach, at least in the context of
1466 an involuntary vigilance-retrieval task administered within the same lab session as
1467 film encoding and the visuospatial intervention. In the General Discussion below, we
1468 consider the wider implications of these findings.

1469 **General Discussion**

1470 The present research was motivated by intriguing evidence that involuntary
1471 and voluntary memory could be functionally dissociated. In two studies led by Lau-
1472 Zhu and two of the present co-authors (Lau-Zhu et al., 2019, 2021), involuntary
1473 memory of a trauma film was decreased in the group who had undergone a
1474 visuospatial intervention (film reminder cues followed by engaging in a visuospatial
1475 task), while voluntary memory for the same material was not affected. These studies
1476 were conducted to follow up on conceptually similar findings, where similar

visuospatial interventions decreased diary-reported intrusions but not voluntary memory for the same material (Asselbergs et al., 2024; Deeproose et al., 2012; Holmes et al., 2009; James et al., 2015). Collectively, such results supported the DRA, a theory that challenged unitary accounts of episodic memory, such as the eCMR developed by another present co-author (Talmi et al., 2019). We formed an adversarial collaboration to explore this issue together, adopting state-of-the-art, resource-intensive, stringently controlled methodology, which involved in-person testing of over 400 research volunteers.

Experiment 1 was carefully designed to test four hypotheses derived from the DRA, and sharpened through extensive discussion with reviewers on the PCI panel (<https://rr.peercommunityin.org/articles/rec?id=807>). The data did not support any of these hypotheses. The failure to support Hypotheses A and B stemmed directly from the failure to replicate the reported effect of a visuospatial intervention on involuntary memory. In Experiment 2, we refined our methodology but again failed to replicate the expected interference effect. Given the robust evidence that participants in these experiments experienced intrusions, proponents of the DRA would suggest that the trauma film encoding was quasi-pathological (Brewin & Field, 2024). In this situation, the DRA predicts further that voluntary memory for the trauma film would be impaired relative to memory of a neutral film (Hypothesis C), but we failed to support that hypothesis as well. Similarly, the DRA would suggest that involuntary and voluntary memory of a trauma film should be uncorrelated because they rely on separate memory traces (Hypothesis D); yet we found clear evidence against this prediction, with the two forms of recall being highly positively correlated. Taken together, our results appear to challenge the DRA. The similarity between the effects

of experimental manipulations on voluntary and involuntary memory instead aligns more closely with unitary accounts of episodic memory.

In the discussion to follow, we will consider in greater depth the context, implications and limitations of each of these findings, before engaging in a broader evaluation of both the DRA and other contemporary theoretical accounts of traumatic memory processes, informed by the present findings.

Hypotheses A and B: Visuospatial Interference Did Not Impact Subsequent Involuntary or Voluntary Memory of an Analogue Trauma or Neutral Film

While the present study aimed to test the *selectivity* of the visuospatial interference effect, it instead produced consistent evidence against an effect of visuospatial interference on any of the forms of memory measured here. The absence of effects on voluntary memory was expected based on similar findings across 11 previous experiments (Asselbergs et al., 2024; Deeproose et al., 2012, Experiment 1; Holmes et al., 2009; Holmes et al., 2010, Experiments 1 and 2; James et al., 2015, Experiments 1 and 2; Kessler et al., 2020; Lau-Zhu et al., 2019, Experiments 1 and 2; Lau-Zhu et al., 2021), but when we formed the null hypothesis for a main effect of the visuospatial intervention, we considered that matching the voluntary and involuntary conditions more tightly here by measuring both through the VRT could result in an effect of the visuospatial intervention on the voluntary measure as well. This possibility had some precedent in the literature, as unlike the aforementioned studies, Hemi et al. (2023) had found that visuospatial interference led to both reduced intrusive memories and lower scores on recognition memory of a trauma film. By contrast, the absence of effects on not only voluntary, but also

involuntary memories here, across our two large-scale experiments, was not predicted.

Considering the diary measure, the majority of lab-based evidence employing a visuospatial intervention has found effects on diary reports of involuntary memories in the 7 days following the exposure to trauma analogue material (see Varma et al., 2024, for a meta-analysis of 11 experiments), but several studies have not (Asselbergs et al., 2018; Brühl et al., 2019; Brennen et al., 2021; Broughill et al., 2025), with one only finding intrusion reduction effects on short-term but not longer-term measures of intrusive memories (Wessel et al., 2025) and another only finding intrusion reduction effects with one visuospatial task but not a different visuospatial task (Matura et al., 2025). In some cases, null results might be attributed to observable deviations from the original protocol (i.e., Holmes et al., 2009), such as use of a visuospatial task that differed in critical ways (Asselbergs et al., 2018), the omission of film reminder cues prior to the visuospatial task (Brühl et al., 2019), or the use of a visual vigilance task as the control intervention, rather than using a modality different from the visuospatial interference task (Wessel et al., 2025). In other cases, it is less clear, and one can only speculate about confounders. For example, it might be that providing a sizeable payment before (rather than after) completing the diary leads to poor diary adherence (Brennen et al., 2021), that unsupervised administration of the intervention online is far less reliable than in-person (Broughill et al., 2025), or that including a structured break itself decreases intrusions (Hørlyck et al., 2019). Our Bayesian evidence against an intervention effect in the diary in both of the present experiments is, however, less likely to be attributable to the above methodological deviations, particularly in the context of our second experiment, which was designed to closely match Lau-Zhu et al. (2019).

Nonetheless, subtle but potentially important methodological differences between that experiment and present work remained, as discussed below.

Even if the lack of an intervention effect on diary intrusions in the present experiments might owe to methodological differences in the diary, the lack of an intervention effect on VRT intrusions is harder to explain. Such VRT intervention effects have been reported in three previous experiments (Lau-Zhu et al., 2019, Experiments 2 & 3; Lau-Zhu et al., 2021), which we aimed to replicate closely here in two studies with greater statistical power (per group $n = 30$ or 60 vs per group $n = 18$ or 19). Aligning with the present findings, recent research from deBettencourt et al. (2026) using an AI-guided implementation of this visuospatial intervention similarly found that, relative to a podcast control condition, the intervention had no effect on intrusions within a same-day involuntary VRT, despite reducing intrusions in a subsequent diary measure. Notably, their version of the VRT did not require memory reports, suggesting that the intervention is unlikely to decrease VRT intrusions regardless of the extent of reporting requirements. Taken together, these data provide compelling evidence that, despite initially promising findings from Lau-Zhu and colleagues, a same-day involuntary VRT does not show effects of visuospatial interference.

Considered more broadly, the present findings evidently call into question the replicability and possible boundary conditions of lab-based research employing visuospatial interventions in the context of analogue trauma in healthy participants and underscore that effects may be fragile and hard to replicate. We now consider possible boundary effects and speculate on their potential implications on diary measures in this study, noting the limitation that the following suggestions are post-hoc. First, since the diary here was not our main focus and to reduce burden, it was

1575 briefer (across four days with one prompt a day, rather than across seven days with
1576 multiple prompts per day) and administered online rather than pen-and-paper, which
1577 might have reduced sensitivity. However, most intrusions occur in the first days after
1578 watching the film and we recorded a similar trajectory of intrusion frequency as in
1579 previous studies (e.g., James et al., 2015). Second, Lau-Zhu and colleagues
1580 administered a single VRT between the intervention and the diary instructions, while
1581 in Experiment 1 we administered two. This took approximately 40 minutes, 3.5 times
1582 longer than the intervention itself. This is unlikely to be the cause of the null
1583 intervention effect, since a recognition task of similar length was administered in Lau-
1584 Zhu et al. (2021) and Experiment 2 replicated the null effect although only a single
1585 VRT was administered.

1586 A more likely explanation is that the effect of the visuospatial intervention on
1587 the diary is found only when the intervention is not followed by detailed memory
1588 reporting that day. Krans and colleagues have shown that administering a verbal
1589 memory measure post-film (a recognition test for just one part of the film) reduced
1590 intrusions for that part of the film (Krans et al., 2009). This test contained neutrally
1591 formulated items intended to rehearse information from the film in the form of
1592 statements that participants had to verify as true or false, such as “There were three
1593 doctors in white coats present at the scene”. The authors suggested that by
1594 performing this memory test immediately after the trauma film, and providing a
1595 memory rehearsal opportunity, these parts of the memory became more integrated,
1596 less isolated and less fragmented, and therefore less likely to intrude. Although
1597 memory reporting requirements do not always have this effect (Lau-Zhu et al., 2019,
1598 2021), Krans et al.’s results suggest that reflecting on trauma-analogue memories
1599 within the VRT, may have inadvertently overridden the attenuating effect of the

intervention task in the experiments reported here, obfuscating differences between conditions in the intrusion diary.

While the experiments which inspired our collaborative research (Lau-Zhu et al., 2019, 2021) each observed effects of the visuospatial intervention on diary intrusions, it is possible that apparently subtle differences in the methodology, described in Supplementary Table 1 and Appendix, caused our participants to provide more detailed memory reports, which attenuated the effects of the visuospatial intervention. We speculate that the unconstrained box which captured memory reports, and the ability to type into it (rather than write with a pen) could have played a role. Additionally, Lau-Zhu and colleagues told participants to “briefly describe the content of the image” so that they could “check it is related to the film” (e.g., the clip of an eye surgery versus a rampaging elephant), while we instructed participants instead to “provide a brief description that allows us to understand what you saw in your mind’s eye”. While these statements are similar, the changes we made nevertheless could have altered how participants approached this task. Although we do not have data to support this claim that the Lau-Zhu et al. (2019) Experiment 2 descriptions were shorter, in that study memory reports were written in a small box using pen-and-paper, and likely to have been brief, amounting to just a few words. By contrast, the unconstrained text entry employed here allowed for memory reports of widely varying lengths (for illustrative purposes, “the image of the elephant terrorizing people came to mind. It was trampling everything in its path. It was probably scared and confused”, “elephant”, and “an elephant escaped the circus”). In the Involuntary VRT, text entry length following each memory report key press ranged from 1 to 83 words ($M = 9.37$, $SD = 7.46$) in Experiment 1 and from 1 to 75 words ($M = 8.11$, $SD = 6.69$) in Experiment 2. This speaks to the possibility that

elaboration upon involuntary memories in this task has obfuscated a potential intervention effect in the subsequent diary measure.

Supporting this possibility, in an experiment focused on clinical analogues of the reduction of intrusive memories through an intervention, rather than testing models of memory per se, no intrusion descriptions in the VRT were allowed (deBettencourt et al., 2026). Critically, in line with the possibility that memory elaboration, either in a VRT (here) or in voluntary tests (Krans et al., 2009) may impact intrusions, deBettencourt and colleagues observed decreased intrusions in the diary. Our results thus support the decision, in recent clinical studies, not to prompt intrusive memory descriptions in the daily diaries but rather focus only on intrusion frequency (Ramineni et al., 2023; Beckenstrom et al., 2026). That said, whether the interference effect deBettencourt et al. (2026) observed in the diary is attributable to differences in the implementation of the diary measure (e.g., 7 day duration, rather than 4 here), the use of a more targeted reactivation procedure before the visuospatial task (guided recall of subjective ‘worst moments’ of film, rather than the standardised film reminder cue images employed here), or eliminating memory reporting in the VRT requires additional research. Clearly, such a constraint on involuntary memory reporting does, in the context of experimental research, limit the potential to test for selective interference between involuntary and voluntary memory under controlled laboratory conditions.

In this context, it is natural to consider other means of empirically testing the DRA in this sample. With this in mind, we now discuss the present findings with respect to two distinct predictions derived from the DRA, which did not depend upon the presence of a visuospatial interference effect.

Hypothesis C: Voluntary Recall was Unimpaired by an Analogue Trauma

The present Experiment 1 provided no indication that voluntary memory was reduced in the negative condition relative to the neutral condition, contrary to Hypothesis C. Before considering the implications of this finding, we briefly return to the method by which we derived this hypothesis from the DRA.

Early iterations of the DRA (Brewin et al., 2010) argued that voluntary recall may be impaired in PTSD only for the most intense moments of a traumatic experience: "it may be only certain traumatic moments (hotspots) that are missing from the C-rep". Moreover, these initial accounts argued that mild emotional manipulations (such as those experienced in lab-based studies of emotional memory) may actually enhance rather than impair voluntary memory encoding. However, these accounts agree that C-reps in the hippocampus will be impaired when this material is encoded in a pathological way, something that should impact adversely any memory measure that relies on intact hippocampal associations, such as free recall. Crucially, the DRA suggests that the presence of intrusions could indicate that pathological encoding occurred. We designed the study to enable detection of pathological encoding by employing an experimental analogue of trauma, which has induced involuntary memories in previous work (e.g., James et al., 2016; Lau-Zhu et al., 2019, 2021) and in the present research. Our finding of unimpaired recall of trauma-analogue material in a sample which presented with substantial involuntary memory for the same material therefore directly contradicts these early iterations of the DRA.

Subsequent iterations of the DRA (Bisby et al., 2016, 2018, 2020), based on neuroimaging data in healthy participants, extended its explanatory scope. Specifically, these later extensions of the DRA predicted that even relatively mild

emotional manipulations would down-regulate hippocampal function (Bisby et al., 2016, 2018, 2020). Initial empirical findings supported this prediction: the researchers observed decreased BOLD signal in the hippocampus when healthy participants encoded emotional materials, and behaviourally, such material impaired associative binding, along with reducing coherence of voluntary recall (i.e., impaired C-reps) and enhancing memory of individual sensory elements (i.e., enhanced S-reps).

The present finding of unimpaired voluntary recall under analogue trauma appears to directly contradict this latter conceptualisation of the DRA. Instead, the present finding aligns more closely with other research (Bogdan et al., 2024; de Montpellier et al., 2025), where negative emotion did not impact paired-associate learning when attentional differences at encoding were controlled. Thus, these findings taken together provide compelling counterevidence to the argument that negative emotion impairs voluntary memory encoding via hippocampal down-regulation (Bisby et al., 2016, 2018, 2020). Furthermore, Kobelt et al. (2024), who examined neural activity during the encoding of a trauma film, found that the encoding of materials which subsequently became intrusive was associated with increased, rather than decreased, hippocampal activation, the extent of which positively correlated with amygdala activation. Additionally, while hippocampal activation was decreased during encoding of negative emotional material in Bisby et al. (2018), the nature of the material did not influence subsequent memory effects in the hippocampus in that study, and in Madan et al. (2017). Hence, there is accumulating behavioural and neural evidence that negative emotion impairs neither hippocampal function during memory encoding nor resultant voluntary recall, even in the most direct laboratory analogue of traumatic memory.

The inverse prediction, made in some formulations of the DRA - that the “perception of a moderately stressful or emotionally salient event results in the creation of more enduring C-reps” (Brewin et al., 2010, p. 221) - is also not supported by the present data, as there was equally no credible Bayesian evidence of enhanced recall of the trauma film relative to the neutral film. Despite using the most intense emotional material that is ethically permissible, we did not observe emotionally enhanced memory. As noted in the Discussion section of Experiment 1, this absence of an effect of emotion on recall was predicted by a unitary computational account of emotional memory (Talmi et al., 2019). According to this account, emotionally enhanced memory is due to competition between emotional and neutral stimuli during retrieval, and therefore not typically observed in between-list (here, between-participant) designs when stimuli are matched on semantic similarity (Ack-Baraly et al., 2016; Talmi, 2013).

To summarise then, the present data provided no evidence that an analogue trauma either impairs or enhances the encoding of (hippocampal-dependent) voluntary memory, as reflected by the number of unique details recalled in the voluntary VRT. This finding not only contradicts the claims of recent extensions of the DRA but provides support for unitary alternatives.

Nonetheless, other approaches to quantifying voluntary recall may have yielded different results. Indeed, future research may consider examining the coherence of voluntary recall and/or the quantity of spatiotemporal vs. item-level details recalled, given that Bisby and colleagues point to these as key aspects of voluntary memory which negative emotion may impair (Bisby et al., 2016, 2018, 2020).

Hypothesis D: Involuntary Recall Predicted Voluntary Recall

By measuring voluntary memory immediately after measuring involuntary memory, using a matched testing format, Experiment 1 also allowed us to examine more carefully whether these two forms of memory retrieval derive from separate traces rather than a single, shared episodic memory representation. Whereas the DRA would be consistent with a lack of correlation, we found strong positive within-subject correlations between involuntary and voluntary tests, across both DV1 (i.e., the total number of memories) and DV2 (i.e., the number of unique details within these memories).

As reviewed in Brewin (2014, Table 3), the majority of previous trauma film paradigm studies seeking to evaluate the correlation between voluntary and involuntary memory measures have not detected an association between them (e.g., Holmes et al., 2004). While clinicians are sensitive to the burden on patients, which is related to the number of intrusions they experience, even if the content of those intrusions is the same every time, mainstream memory researchers quantify voluntary memory through the number of unique items that participants remember. This leads to a divergence between the two research traditions in the measures they employ. We have reviewed extensively Lau-Zhu et al. (2019, Experiment 2) where voluntary and involuntary memory were measured using DV2 and DV1, respectively; more typically, voluntary memory is measured using a recognition memory test, where the hit rate is a count of the number of studied items that participants recognise as “old” (e.g., Hørlyck et al., 2019, Experiment 1; Lau-Zhu et al., 2021). It is possible, therefore, that previous research may have failed to identify the associations observed here due to unaddressed differences in the measurement of involuntary and voluntary memory, both with respect to differing testing formats

(diary vs recognition or recall) and differing dependent variables being examined (counts vs details). Indeed, even measures of voluntary memory are often “stochastically independent” because of these processing differences, which is why such findings are considered only limited evidence for the presence of separate memory systems (Schacter, Wagner & Buckner, 2000).

Some previous research has also observed positive correlations between measures of involuntary and voluntary memory of a trauma film (Ferree & Cahill, 2009; Ferree et al., 2011). Importantly however, these studies shared two methodological weaknesses: firstly, they measured intrusions through retrospective estimates of intrusion frequency rather than direct reporting of each intrusion, and secondly, this measure of intrusions was delivered after a voluntary memory test. As Brewin (2014) reasonably argues, this means that intrusion estimates may be biased by the content of voluntary recall which immediately preceded them. By contrast, our experimental design mitigates these issues, and provides the most compelling evidence to date that these two forms of recall are indeed positively associated, a finding best aligned with unitary accounts (Berntsen, 2021; Cohen & Kahana, 2022; Rubin et al., 2008; Talmi et al., 2019). That said, a positive correlation could be an artefact of other individual differences, such as greater motivation in some participants than others, which would lead them to engage more with the film and focus more on both retrieval tasks.

Theoretical Implications

The comprehensive lack of support for the DRA in the present study raises important questions both regarding the generalisability of this account and the potential promise of unitary alternatives. We found evidence against selective

1775 interference, emotion-induced voluntary memory impairment, and uncorrelated
1776 involuntary and voluntary memory traces. Collectively, our findings raise
1777 considerable doubt regarding pre-existing evidence for the DRA derived from
1778 laboratory studies in healthy participants. Based on this collection of findings, we
1779 argue that the DRA has been overextended, from its initial status as a theoretical
1780 account of clinical PTSD (Brewin et al., 1996, 2010; Brewin, 2014), to a wide-
1781 reaching account of emotional memory processes (Bisby et al., 2016, 2018, 2020;
1782 Hørlyck et al., 2019), evoked, for example, to explain phenomena such as the effect
1783 of emotional images on temporal memory (reviewed in Talmi & Palombo, 2025). Our
1784 results do not bear on the question of whether the DRA accounts for the clinical
1785 condition of PTSD, which it was originally conceptualised to explain (Brewin et al.,
1786 1996, 2010; Brewin, 2014).

1787 There are clear implications of the present study with respect to future work
1788 seeking to experimentally evaluate the DRA via investigation of selective
1789 interference. While potential boundary conditions, discussed above, suggest caution
1790 in interpreting the null effect of selective visuospatial interference on the diary
1791 measures, the balance of evidence indicates that such an effect in healthy
1792 participants is not reliable using a same-day lab-based measure of involuntary
1793 memory (VRT). We recommend, therefore, that future research on the selective
1794 interference effect returns to real-life diaries to measure involuntary memory. It is far
1795 harder to control diary reports of involuntary memory so that they are fully
1796 comparable to voluntary memory measures, rendering a stringent test of the DRA
1797 more challenging.

1798 Our research has addressed involuntary memories but not dissociative
1799 flashbacks, which are difficult, if not impossible, to study in healthy samples.

Findings from contemporary clinical neuroscience suggest that dissociative flashbacks in PTSD may rely on distinct, extra-hippocampal, forms of memory representation (Kearney & Lanius, 2024), perhaps a form of conditioned response (Milton, 2019).

We hope that the research reported here will encourage investigations that test predictions of formal unitary models for highly emotionally intense experiences that are not traumatic. Such work can help develop those unitary models that followed the tradition of the Temporal Context Model (Howard & Kahana, 2002), and address limitations that challenge their application to the recall of such experiences. At present, these models do not account well for the role that the intention to remember plays in the retrieval process, so they may struggle to account for any differences between voluntary and involuntary memory (Berntsen, 2023). Additionally, they do not offer a ready mechanism to explain the influence of goals on the retrieval process, such as a goal to suppress traumatic memory to protect mood (Mamat et al., 2024) or to maximise accuracy in a court of law. While retrieved-context models have successfully accounted for temporal dynamics of recall even outside the lab, their evidence base comes mostly from intentional recall of words, so more evidence is needed to show that they provide the best explanation for one-shot learning of meaning-rich, sensory material, such as can occur after traumatic experiences. We note also that in Experiment 1 there were, surprisingly, more lab-based involuntary memories of the neutral film than the traumatic film, suggesting that it may be feasible to study involuntary memory in healthy samples using exclusively neutral stimulus material. Future research could start from simulations using retrieved-context models to define cases that best test model assumptions.

1825 **Constraints on Generality**

1826 The participants in this research were all healthy adults, screened specifically
1827 to ensure no prior history of, or heightened vulnerability to, psychological trauma.
1828 Owing to the experiments being conducted in-person at the University of Cambridge,
1829 these participants were also likely to have been predominantly from western,
1830 educated, industrialized, rich, and democratic (WEIRD) backgrounds (Henrich,
1831 Heine, & Norenzayan, 2010). These factors limit the generalisability of the present
1832 findings to distinct sample populations, including children and adolescents, as well
1833 as individuals from non-WEIRD cultures, and individuals experiencing trauma-related
1834 clinical conditions such as PTSD.

1835 These limitations were deemed justifiable, to ensure that this research could
1836 be conducted in compliance with ethical guidelines, using rigorously controlled in-
1837 person experiments, and could produce findings which were directly comparable with
1838 previous laboratory-based research in this area.

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Declaration of Competing Interests

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Appendix

Vigilance Retrieval Task (VRT) Instructions Across Experiments

Present Experiment 1 Voluntary VRT	Present Experiment 1 Involuntary VRT	Present Experiment 2 Involuntary VRT	Lau-Zhu et al. (2019) Experiment 2 Involuntary VRT
At the same time that you are doing the number task you will also see pictures in the background, but you don't have to do anything with them. Continue to focus on the number task.	At the same time that you are doing the number task you will also see pictures in the background, but you don't have to do anything with them. Continue to focus on the number task.	At the same time that you are doing the numbers task you may also see pictures in the background, but you don't have to do anything with those pictures. Instead, continue to focus on the numbers task.	At the same time, on some trials you will also see a background image, but you don't have to do anything with these. Just focus on the number task. I will walk you through these so you know what it looks like and how you should do the task.
We are interested in knowing what you can deliberately recall from the films that you saw. The number task could be quite boring. Please make an effort to recall the content of the film while also trying to focus your attention on doing the number task.	Because this task could be quite boring, you might find that occasionally your attention wanders. An involuntary memory from the film, in the form of a mental image, may pop into your mind, out of the blue, and not at times when you are deliberately thinking about the film.	Because this task could be quite boring, you might find that occasionally your attention wanders. In particular, an involuntary memory from the film, in the form of a mental image, may pop into your mind, out of the blue, like a snapshot from a film scene.	Now, because this task could become quite monotonous, you might find that occasionally your attention might wander off. An mental image from the film may pop into your mind spontaneously, out of the blue. We call this an intrusive image-based memory.
The memories you recall might be mental images that include shorter or longer segments of scenes from the film. They can be vivid and clear, and very detailed, as if you're watching the film again, or quite fuzzy and brief, like a quick flash or blurred snapshot from a scene.	Mental images take the form of pictures in your mind's eye, that is, picturing a scene from the film. Although mental images can sometimes be vivid and clear, as if you're in the film again, they can also be quite fuzzy and brief, like a quick flash or blurred snapshot from a scene.	Mental images are like seeing something in your mind's eye, such as seeing a moment from the film again. Mental images are typically quite fuzzy and brief, like a quick wisp or flash from a scene, but they can sometimes be vivid and clear.	Although imaged-based memories from the film can be extremely vivid and clear, they can also often be quite fuzzy and quite brief, like a quick flash.
To repeat, by deliberately trying to remember we mean that you decide to recall the memory and make an effort to recall it.	To repeat, by involuntary we mean that the mental image pops into your mind spontaneously, rather than you deliberately trying to remember it.	To repeat, by involuntary memories we mean that these mental images pop [gesture!] into your mind spontaneously, rather than you	We are interested in all kind of intrusive image-based memories from the film only. Is that clear?

Is that clear?	We are interested in knowing about the involuntary memories of the film, which you have seen earlier. Is that clear?	deliberately trying to remember them. We are interested in knowing about any such involuntary images from the film that you see in your mind's eye during this task. To check that is clear, could you please explain, in your own words, what involuntary memories are?	
I want you to press the 'SPACE' key every time you manage to deliberately recall a film-based memory. You should press it each time you deliberately recall a memory, even if it's a memory you already recalled before. Please continue trying to deliberately recall memories from the film throughout the numbers task.	I want you to press the 'SPACE' key every time you notice an involuntary memory popping into your mind during the task. You should press it each time you notice an involuntary memory, even if it's the same memory you've had before. But if something doesn't come to mind, don't worry; there's no need to deliberately search for anything.	Every time you notice an involuntary memory popping into your mind during the task, I want you to press the 'SPACE' key. You should press it each time you notice an involuntary memory, even if it's the same one appearing again. But if something doesn't come to mind, don't worry; there's no need to deliberately search for anything.	Whenever an image-based memory from the film pops up, I would like you to complete one row of this sheet. First you will have to briefly describe the content of the image so we can check it is related to the film. Then, you will have to provide 3 ratings on a 5 point scale. First, how distressing was the image? Second, how vivid was the image? Finally, to what extent did you feel you were seeing the film now? I want you to press the '2' key every time you notice an intrusive image of the film popping into your mind during the task.
Once you press the 'SPACE' key, you will see a different screen, prompting you to report details of your deliberate memory from the film. Describe what you see or hear in your deliberately recalled memory. It's ok to repeat memories you already reported. Please don't worry about spelling or grammar. Once you are done reporting, press the 'enter' key to continue the task. Is that clear?	Once you press the 'SPACE' key, you will see a different screen, prompting you to report details of your involuntary memory from the film. Describe what you see or hear in your involuntary memory. It's ok to repeat memories you already reported. Please don't worry about spelling or grammar. Once you are done reporting, press the 'enter' key to continue the task. Is that clear?	Once you press the 'SPACE' key, you will see a different screen, prompting you to describe your involuntary memory from the film. Please provide a brief description that allows us to understand what you saw in your mind's eye. Please don't worry about spelling or grammar. Once you are done reporting, press the 'enter' key to continue the task. Remember that it's ok to repeat memories you already reported.	The screen will go blank and you will have to write down a few details about the image that you just experienced. Let's walk through an example together. You should press the key each time you notice an intrusive image from the film, even if it's the same image you've had before. But if something doesn't come to mind, there's no need to search for anything. Is that clear? Remember: press the GO key whenever you see a digit, unless it is a 3; stop the task whenever you notice an

			image from the film popping into your mind spontaneously. Any questions before we start with the task?
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2349 *Note.* The instructions listed here for Lau-Zhu et al. (2019) Experiment 2 were

2350 extracted from Lau-Zhu's (2016) doctoral thesis, following confirmation that the

2351 thesis reflects the complete procedure (Lau-Zhu, personal communication, 2026).

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Supplementary Materials

Supplementary Table 1

Detailed Methodological Comparisons Between Lau-Zhu et al. (2019) Experiment 2 and the Present Experiments 1 and 2

Methodological Aspect	Lau-Zhu et al. (2019) Experiment 2	Present Experiment 1	Present Experiment 2
Films and Film-Related Measures			
Example videos during consent	No example videos shown before trauma film itself.	Two example videos shown before signing consent form, featuring similar content to trauma film.	Two example videos shown before signing consent form, featuring less similar content to trauma film.
Film viewing set-up	Film displayed on a desktop screen.	Film displayed on a wall by a projector.	Same as present Exp1.
Film viewing instructions	Emphasised attention and immersion into the film.	Same as Lau-Zhu et al. (2019) Exp2.	Same as Lau-Zhu et al. (2019) Exp2.
Trauma film content & duration	11 clips; 12 minutes.	Same as Lau-Zhu et al. (2019) Exp2.	Same as Lau-Zhu et al. (2019) Exp2.
Film as independent variable	All participants saw the trauma film.	Experimenter blinded to film allocation (trauma or neutral).	Same as Lau-Zhu et al. (2019) Exp2.
Mood ratings	Pre-film, post-film and post-intervention.	Same as Lau-Zhu et al. (2019) Exp2.	Same as Lau-Zhu et al. (2019) Exp2.
Other immediate post-film questionnaires	Immediately after film viewing, participants rated film-induced distress, attention to the film and self-relevance of the film using 0-10 scales.	Immediately after film viewing, participants rated film-induced distress, attention to the film, immersion and percentage of time looking away from the film using 0-100 scales.	Same as present Exp1.
Film clip ratings	None.	At the end of the lab-based session, participants saw a title and still from each of the 11 clips they saw and rated their impact, valence and arousal.	Same as Lau-Zhu et al. (2019) Exp2.
Visuospatial Intervention and Control Condition			
Visuospatial task practice	Prior to film viewing. Instructions for Tetris with mental rotation then practice for 3 min.	After reminder cues. Instructions for Tetris with mental rotation then practice for 1 min.	Prior to film viewing. Instructions for Tetris with mental rotation, including an example video and

			comprehension checks, then practice for 3 min.
Filler task, between film viewing and intervention	30 min. 10 min answering a question sheet about a non-fiction book, 10 minutes rating pleasantness of classical music excerpt, then another 10 minutes of the book task.	5 min. Answering a question sheet using a thesaurus.	Same as Lau-Zhu et al. (2019) Exp2.
Film Reminder cues	1 frame per film clip, 3s presentation each; presented in darkened room with instructions to pay attention.	Same as Lau-Zhu et al. (2019) Exp2.	Same as Lau-Zhu et al. (2019) Exp2.
Blinding to intervention allocation	Experimenter blinded to intervention until its administration.	Same as Lau-Zhu et al. (2019) Exp2.	Experimenter blinded to intervention throughout the experiment.
Visuospatial condition after the reminder cues	After the reminder cues, participants engaged in the visuospatial task for 10 minutes.	After the reminder cues, participants completed visuospatial task practice, then engaged in the task for 10 minutes.	After the reminder cues, participants were instructed by a computer program to engage in the visuospatial task for 10 minutes.
Control condition after the reminder cues	Passive control. Participants were instructed to rest for 10 min, with no stimulus to attend to.	Active control. Participants listened to a 10 min podcast about elections and answered simple comprehension questions.	Passive control. Participants were instructed by a computer program to rest for 10 min, with a countdown displayed on screen.
Experimenter presence during the intervention or control conditions	The experimenter left the room and returned when the time ended.	Same as Lau-Zhu et al. (2019) Exp2.	A sound played to indicate the end of the task and the participant was instructed to notify the experimenter that the task had ended.
Demand ratings: expectation regarding the task manipulation on changing intrusive memories	At end of initial lab session.	After film ratings, at end of lab session.	None.
Lab-based Involuntary and Voluntary Memory Measures			
Practice trials of VRT	36 practice trials to familiarise participants with the digit-vigilance component of the task.	Two sets of 12 practice trials (24 total). During the last 12 trials, the participant was asked to	Same as present Exp1.

		press the memory report key and type in the word 'TEST' to familiarise them with this aspect of the task.	
Timing of Involuntary memory test	On day of film viewing and 7 days after film viewing.	On day of film viewing.	Same as present Exp1.
Format of involuntary memory test	VRT with blurred foil or film image displayed every three trials. Participant presses a button each time an involuntary memory occurred.	Same as Lau-Zhu et al. (2019) Exp2.	Same as Lau-Zhu et al. (2019) Exp2.
Involuntary VRT: Instructions regarding intrusions	Used terminology of 'intrusive image-based memories'. See Appendix.	Used terminology of involuntary memories, explained as mental images. See Appendix.	Similar to present Exp 1 but improved for clarity and included further comprehension checks. See Appendix.
Involuntary VRT: Format of inputting intrusions (for full instructions see Appendix)	Press an 'Intrusion' key to pause the VRT and note the content of the intrusion into a small box, using a pen-and-paper sheet. Each intrusion was rated on 5-point scales for distress, vividness and nowness.	Press the space key to pause the VRT and type the content of the intrusion within the same computerised task in an unconstrained box. No further questions about the intrusion.	Same as present Exp1.
Additional lab-based involuntary memory task	A two-minute rest period during which participants were asked to report any spontaneous intrusions of the film, by marking a pen-and-paper sheet.	None.	Same as present Exp1.
Timing of voluntary memory test	7 days after film viewing.	On day of film viewing.	None.
Format of voluntary memory test	Recognition memory test.	Voluntary vigilance retrieval task (VRT). See Appendix for instructions.	None.
Memory test order	Fixed order. Participants were tested for involuntary memory (day 1), then intrusion diary (day 1-7), then involuntary again (day 8), then voluntary memory (day 8)	Order of lab-based involuntary and voluntary memory tests (day 1) is counterbalanced across participants; however, only the first of these is used to test the primary hypotheses. Both were always prior to the diary.	Fixed order. Involuntary VRT (day 1) and then involuntary memory diary.
Involuntary Memory Diary			

Format of intrusion diary	Paper-based. Intrusions reported as they occurred. Definition of intrusion and example on the diary that participants could refer to.	Online questionnaire. Across 4 days. Number and content of intrusions reported in a questionnaire each evening. No definition or examples within diary itself.	Same as present Exp1 except removed references to mental images that were sounds/heard.
Diary structure & duration	7 days (day of study + 6 following days). Each day divided into 3 time periods (morning, afternoon, evening) giving 21 time slots. Diary collected in-person and experimenter verified entries with the participant.	4 days (day of study + 3 following days) Diary prompt sent once per day, giving 4 entry time slots. Diary returned electronically, and entries verified based only on text entered.	Same as present Exp1.
Diary entry length	Constrained by size of box on paper.	Unconstrained.	Same as present Exp1.
Adherence to diary	Diary compliance rating provided by participant.	No rating of diary compliance.	Same as present Exp1.
Contextual Aspects			
Number of lab sessions	Two sessions, one week apart.	One session.	Same as present Exp1.
Experimenter and training	Single PhD student, with direct training from E.A.H. Piloting with feedback, including reviewing ratings, intervention performance, and incoming intrusive memory data.	Multiple research assistants, with training from a postdoctoral researcher trained by E.A.H. Each experimenter piloted the trauma involuntary condition, and was allowed to continue only after intrusive memory data were reviewed by the postdoctoral researcher.	Single postgraduate research assistant, with direct training from E.A.H. Blinded to outcome throughout, so no piloting or reviewing data occurred as part of feedback.

Note. Elements within each section are organised in the order of their appearance in the experimental protocols. The methodological details listed here for Lau-Zhu et al. (2019) Experiment 2 were extracted from Lau-Zhu's (2016) doctoral thesis, following confirmation that the thesis reflects the complete procedure (A. Lau-Zhu, personal communication, 2026).

Supplementary Table 2

Experiment 1 VRT Repetitions and External Details in Each Group

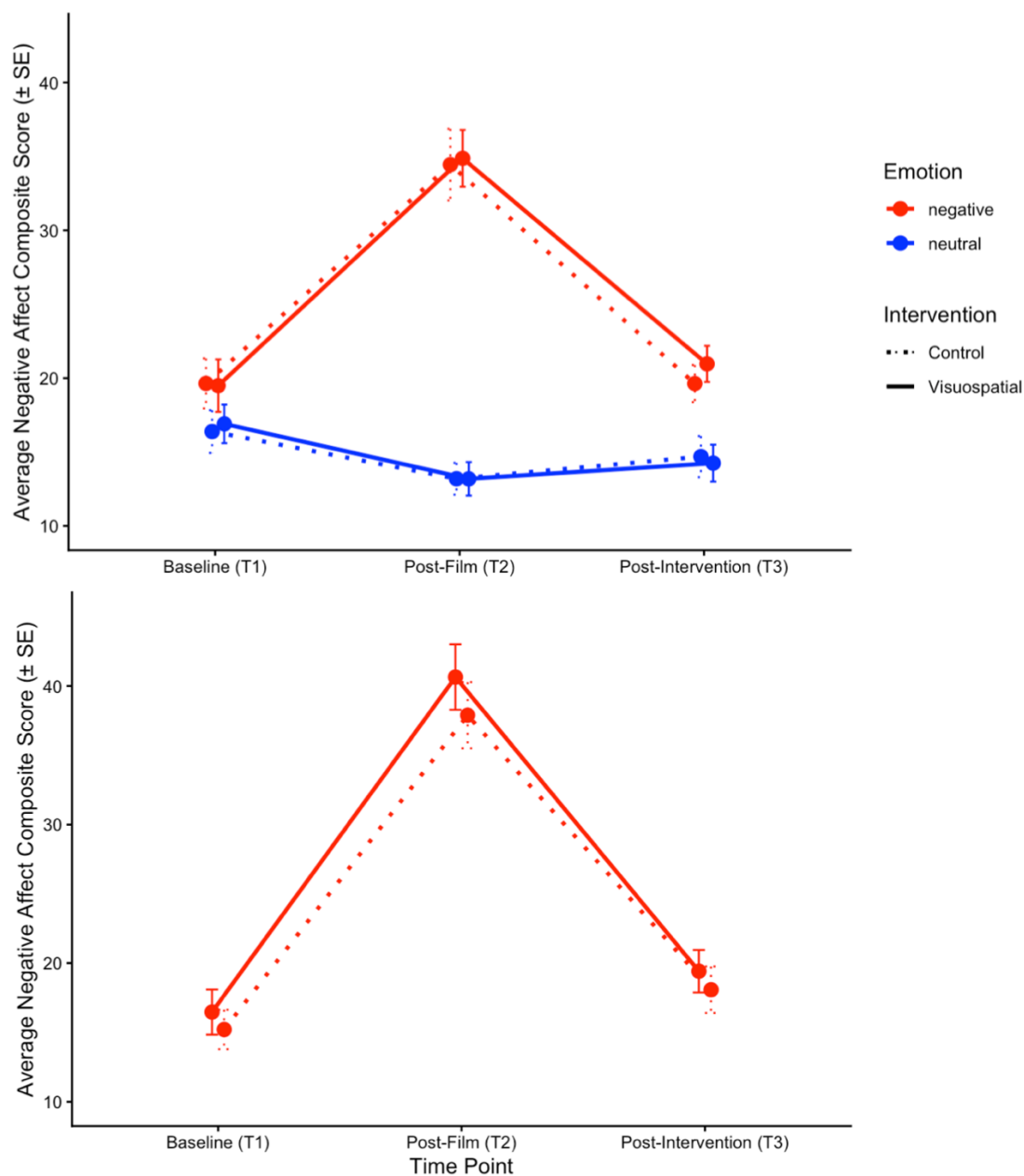
Intentionality	Emotion	Intervention	Mean Repetitions	SD Repetitions	Mean External Details	SD External Details
Involuntary	Negative	Control	4.40	4.32	1.23	2.10
Involuntary	Negative	Visuospatial	4.93	3.90	1.37	2.88
Involuntary	Neutral	Control	6.43	4.64	0.47	1.01
Involuntary	Neutral	Visuospatial	6.33	6.02	0.77	2.40
Voluntary	Negative	Control	8.97	11.59	2.87	6.94
Voluntary	Negative	Visuospatial	7.57	14.82	1.23	2.18
Voluntary	Neutral	Control	8.33	9.83	1.03	2.63
Voluntary	Neutral	Visuospatial	6.97	6.49	1.63	4.73

Note. Repetitions refers to instances in which an Internal detail (i.e. a detail from the film) was repeated within the same memory report (e.g., “the eye surgery” [a unique Internal detail], followed later in the test by “the eye being operated on” [a repetition of the same detail]). External details reflect instances of content in memory reports which did not refer to the film.

Supplementary Figure 1

Negative Affect at Each Time Point in Each Emotion and Intervention Group in

Experiment 1 (Top) and Each Intervention Group in Experiment 2 (Bottom)

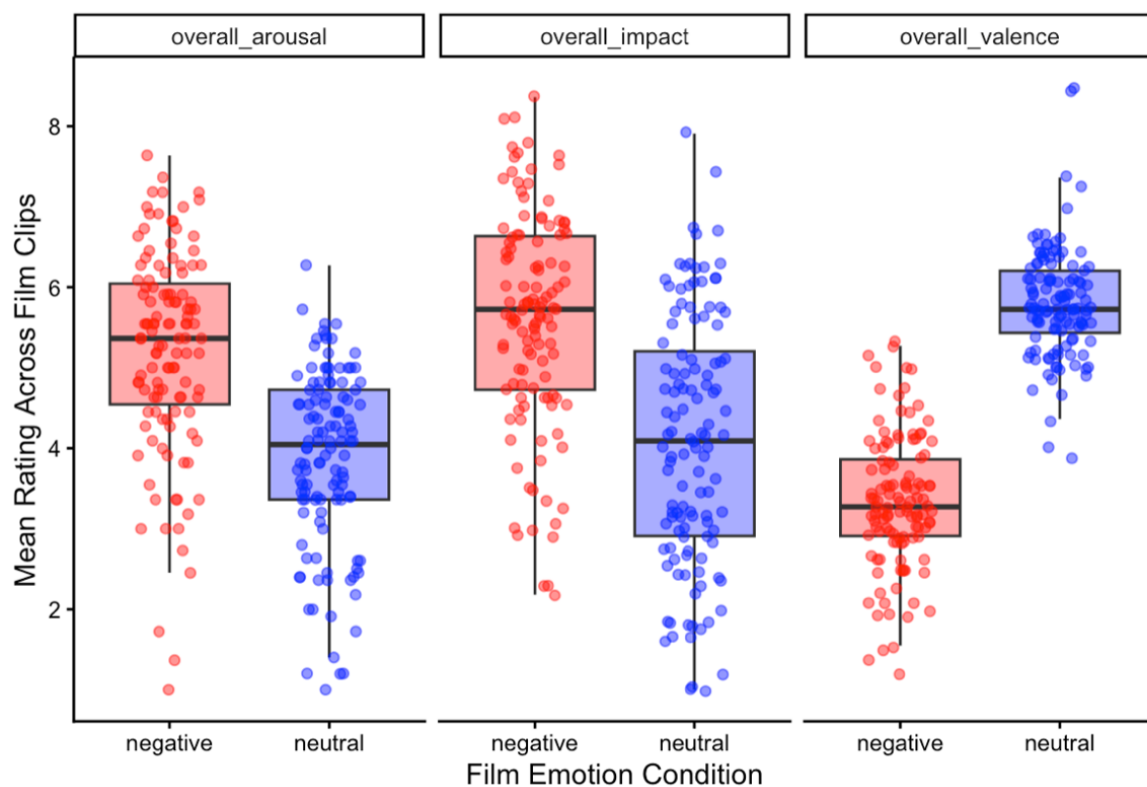


Note. Values were derived from the mood ratings questionnaires administered at 3 time points during the lab session. Points indicate the mean negative affect composite score (see text) for N=60 in each group in Experiment 1 (Top, collapsing

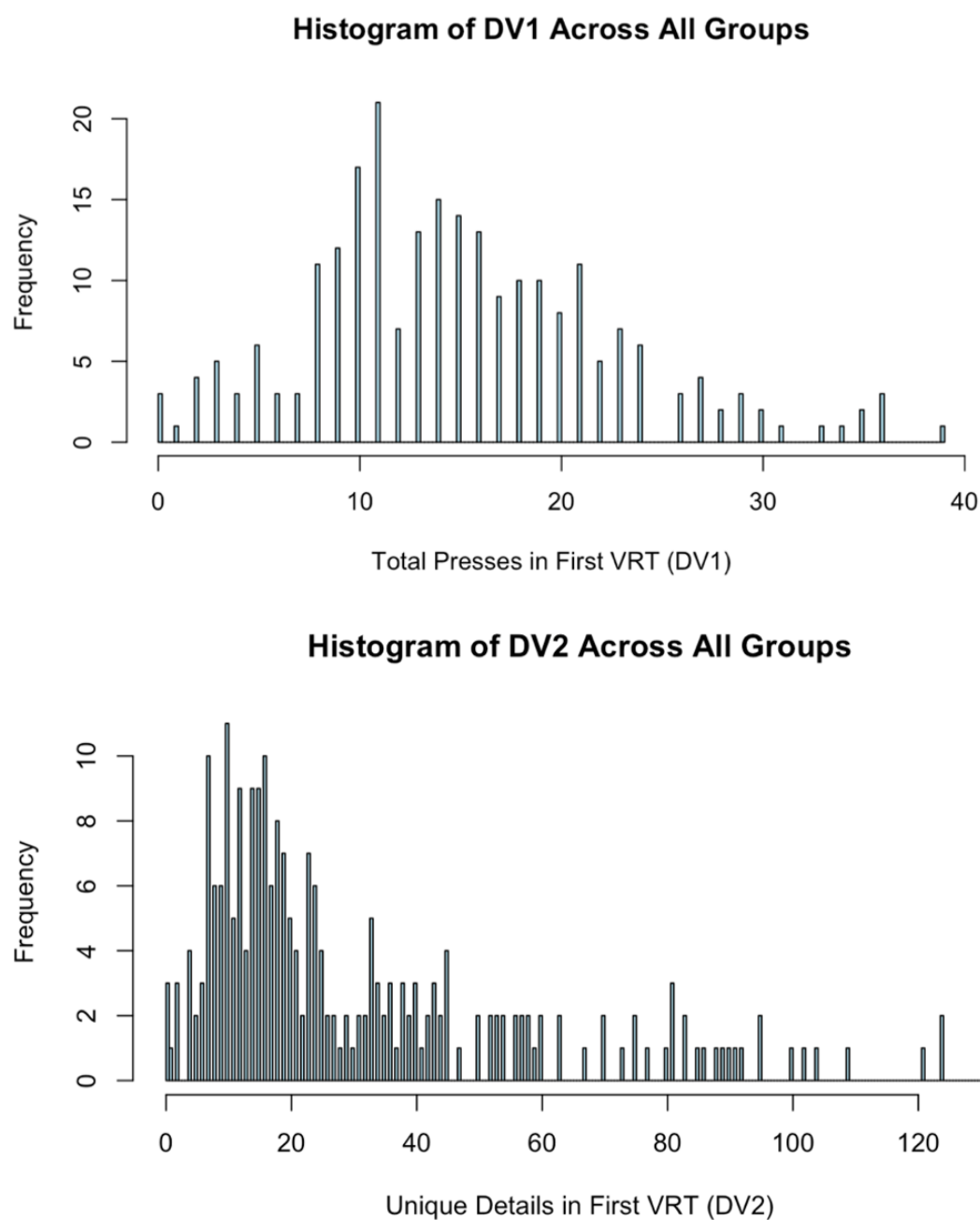
across the intentionality factor) and in Experiment 2 (Bottom). Error bars denote the standard error.

Supplementary Figure 2

Experiment 1 Film Ratings in Each Emotion Condition Averaged Across All Clips



Note: Left: Arousal ratings. Middle: Impact ratings. Right: Valence ratings. Points represent individual average ratings across the 11 film clips viewed. Boxplots indicate the group-level medians and interquartile ranges across the two Emotion conditions (N=120 per group, collapsing across the Intentionality and Intervention condition).

2420 **Supplementary Figure 3**2421 *Histograms of DV1 and DV2 in Experiment 1*

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2424 *Note: Distributions of variables derived from the first Vigilance Retrieval Task (VRT)*2425 *of Experiment 1. Top: DV1, Bottom: DV2.*

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